



9b. Multimodal Learning

Never Stand Still

Faculty of Engineering

COMP9444 Week 9b

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Goal

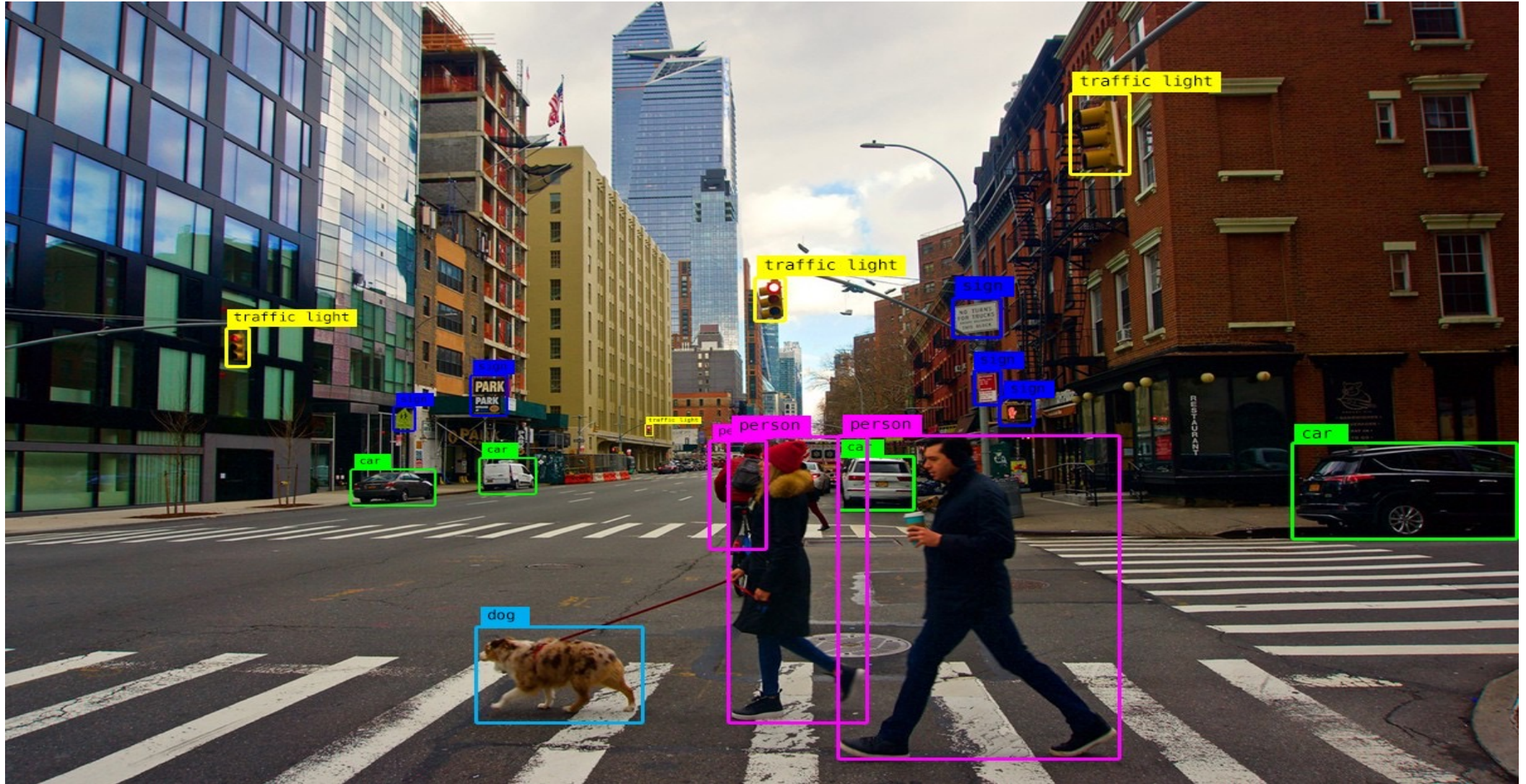
- To motivate the need for “Computer Vision + Natural Language Processing”
- After the talk, everyone can confidently say: “yeah, I know various tasks at the intersection of computer vision and natural language processing”
- Focus on high-level overview, not technical details
- Focus on static images, not videos (although they are easy to translate to videos)
- Focus on selective set of papers for various tasks, not a comprehensive literature review

Agenda

- Computer Vision
- Natural Language Processing
- Computer Vision + Natural Language Processing
- Building Blocks
 - Convolutional Neural Networks (CNNs)
 - Recurrent Neural Networks (RNNs)
 - Attention Mechanism
- Encoder-Decoder Framework
- Image Captioning
- Visual Question Answering (VQA)
- Visual Dialog (VisDial)
- Vision-Language Navigation (VLN)
- Visual Grounding
- Summary

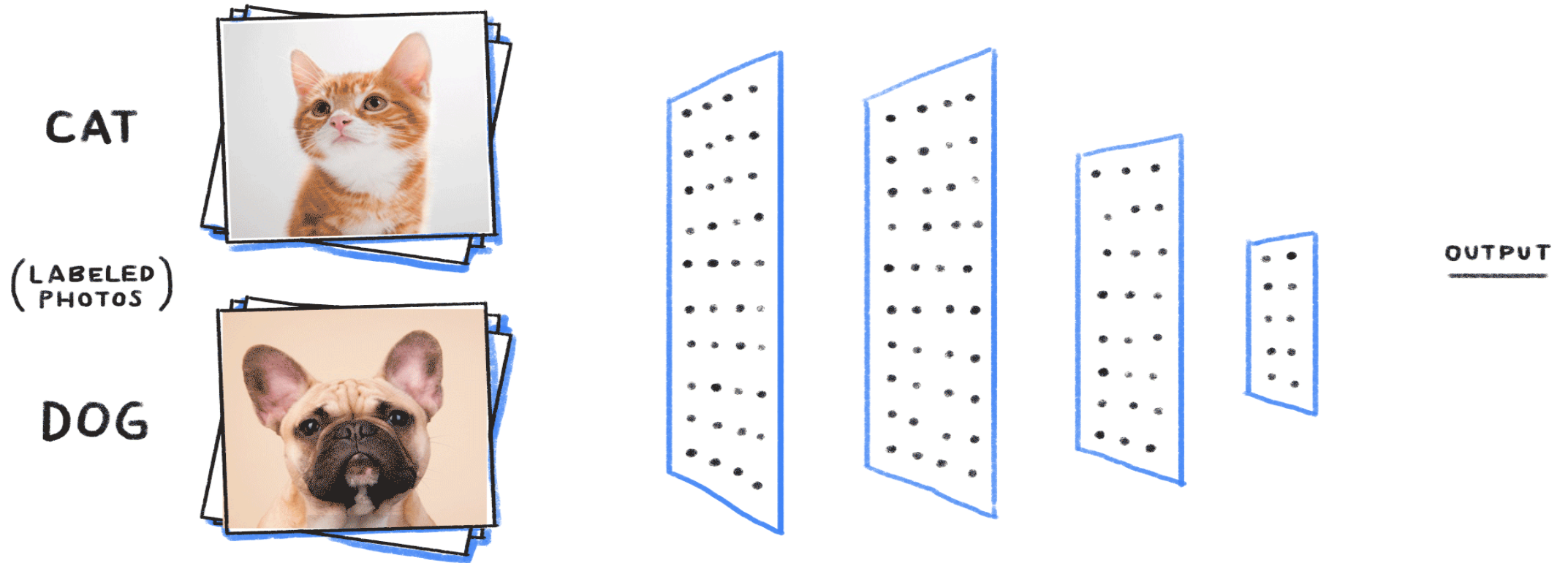
Computer Vision

- Enabling machines to process, represent, understand, and generate visual data



CV Applications

➤ Image Classification



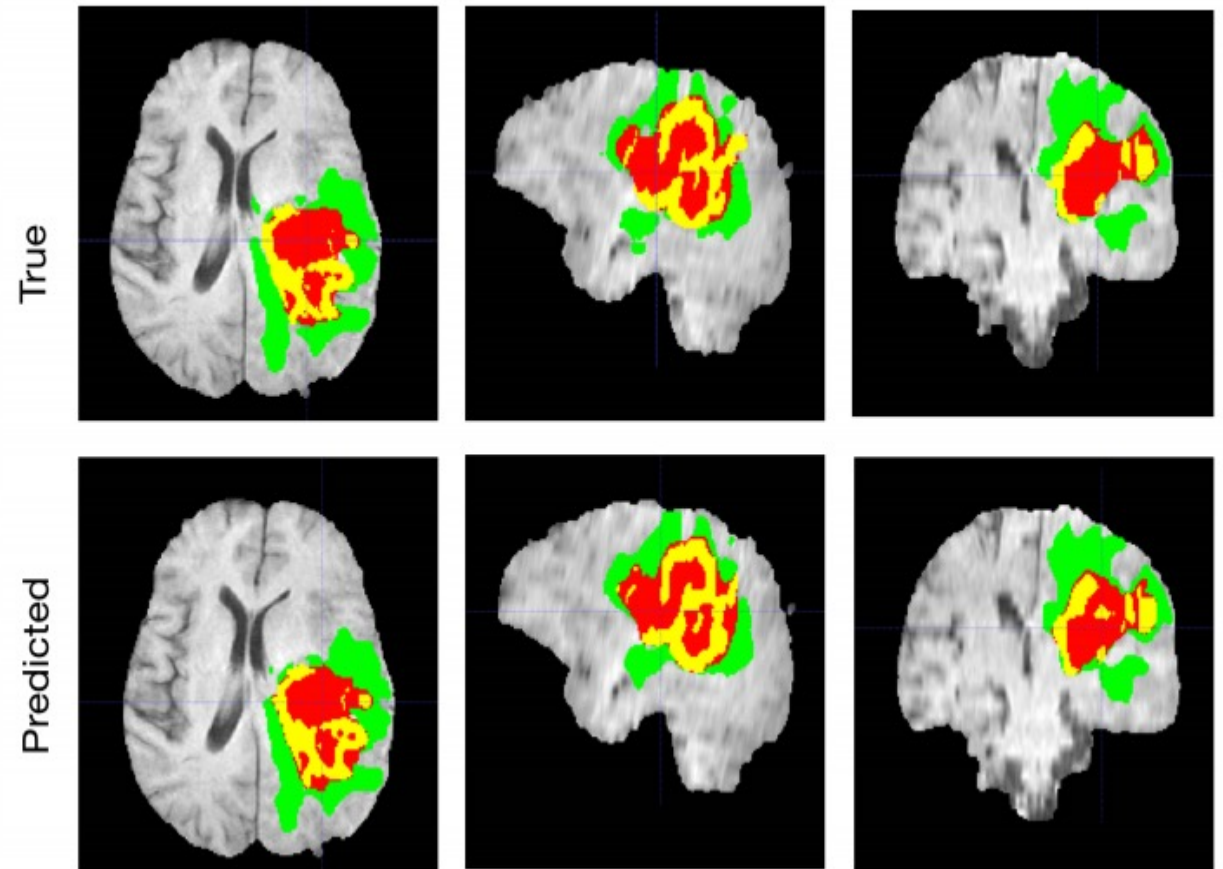
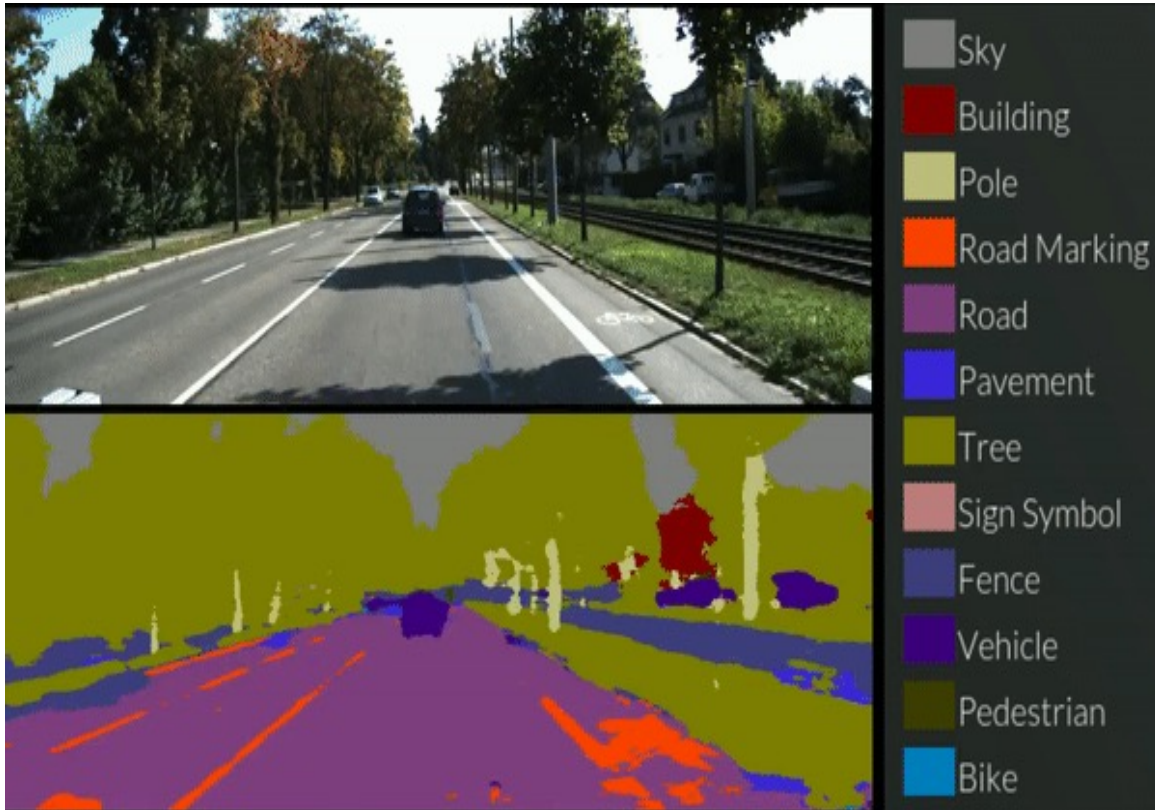
CV Applications

➤ Object Detection



CV Applications

➤ Segmentation



Natural Language Processing

- Enabling machines to process, represent, understand, and generate languages

In fact, the **Chinese** **NORP** market has the **three** **CARDINAL** most influential names of the retail and tech space – **Alibaba** **GPE**, **Baidu** **ORG**, and **Tencent** **PERSON** (collectively touted as **BAT** **ORG**), and is betting big in the global **AI** **GPE** in retail industry space. The **three** **CARDINAL** giants which are claimed to have a cut-throat competition with the **U.S.** **GPE** (in terms of resources and capital) are positioning themselves to become the ‘future **AI** **PERSON** platforms’. The trio is also expanding in other **Asian** **NORP** countries and investing heavily in the **U.S.** **GPE** based **AI** **GPE** startups to leverage the power of **AI** **GPE**. Backed by such powerful initiatives and presence of these conglomerates, the market in APAC AI is forecast to be the fastest-growing **one** **CARDINAL**, with an anticipated **CAGR** **PERSON** of **45%** **PERCENT** over **2018 - 2024** **DATE**.

To further elaborate on the geographical trends, **North America** **LOC** has procured **more than 50%** **PERCENT** of the global share in **2017** **DATE** and has been leading the regional landscape of **AI** **GPE** in the retail market. The **U.S.** **GPE** has a significant credit in the regional trends with **over 65%** **PERCENT** of investments (including M&As, private equity, and venture capital) in artificial intelligence technology. Additionally, the region is a huge hub for startups in tandem with the presence of tech titans, such as **Google** **ORG**, **IBM** **ORG**, and **Microsoft** **ORG**.

NLP Applications

➤ Text Understanding

"I love this movie.
I've seen it many times
and it's still awesome."



"This movie is bad.
I don't like it at all.
It's terrible."



NLP Applications

➤ Machine Translation

Google Translate



Sign in

Text

Documents

DETECT LANGUAGE

ENGLISH

SPANISH

FRENCH



GERMAN

ENGLISH

SPANISH



I love teaching humans and machines



Ich liebe es, Menschen und Maschinen beizubringen ☆



35 / 5000



[Send feedback](#)

NLP Applications

➤ Question Answering/Comprehension

Passage Sentence

In meteorology, precipitation is any product of the condensation of atmospheric water vapor that falls under gravity.

Question

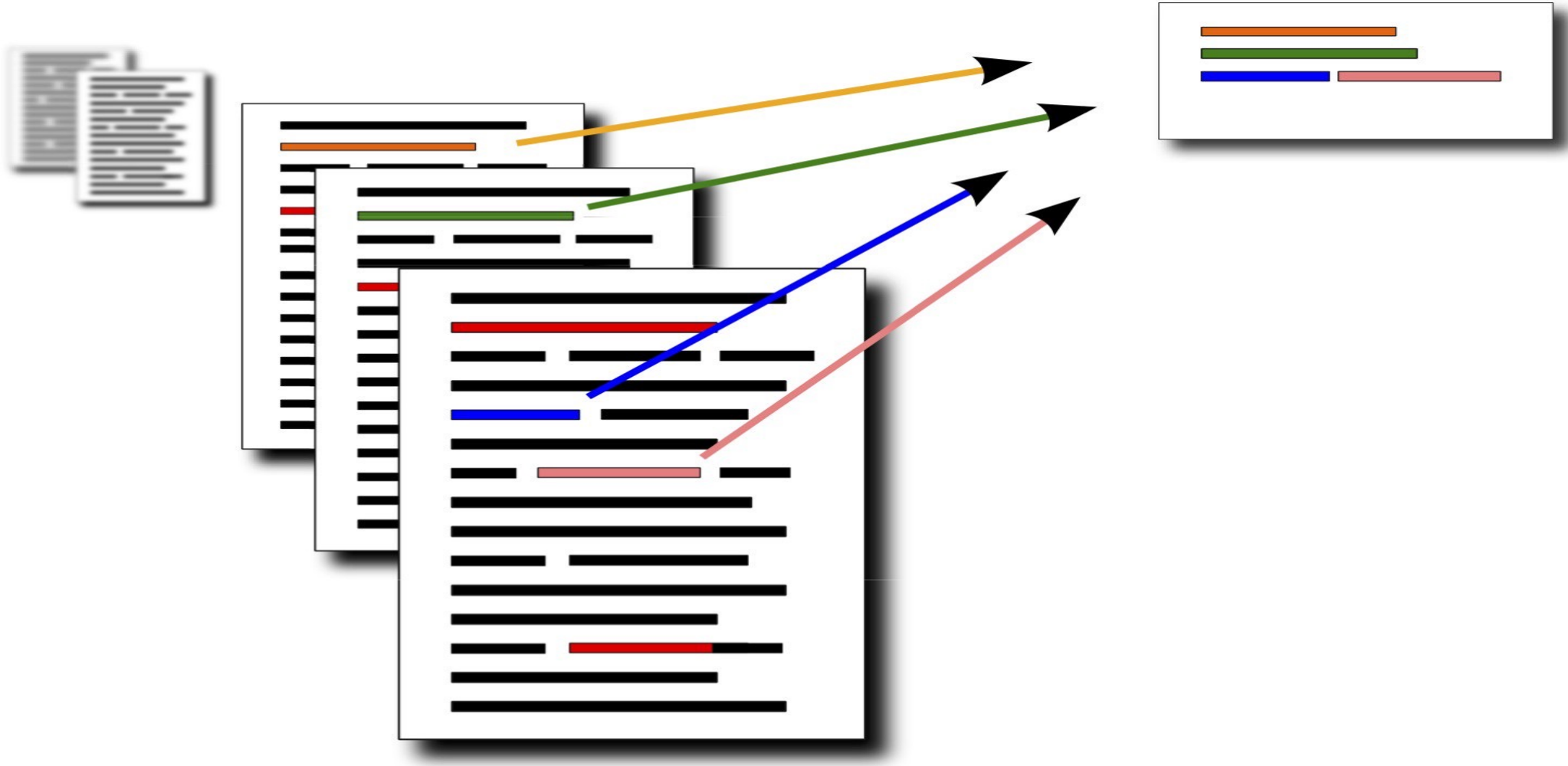
What causes precipitation to fall?

Answer Candidate

gravity

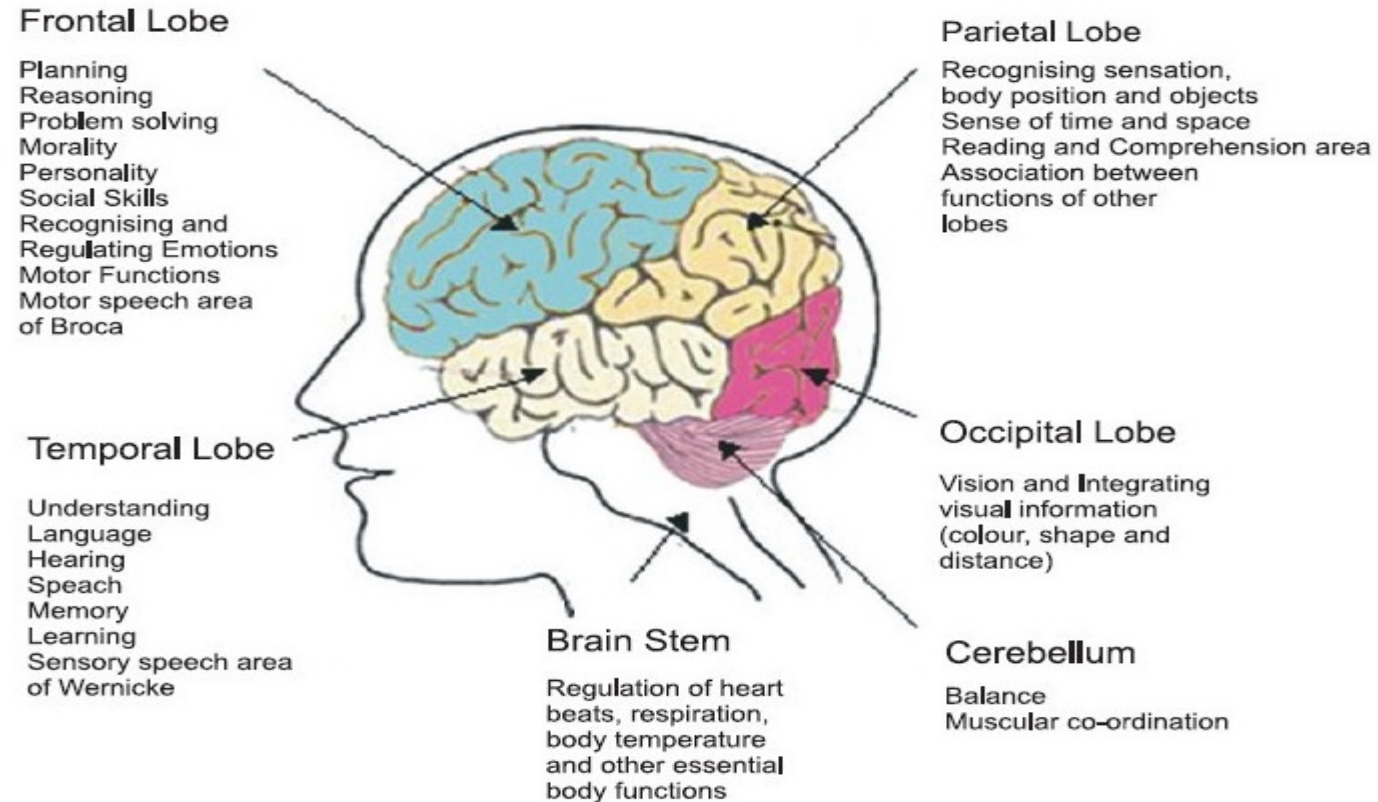
NLP Applications

➤ Text Summarization



Vision + Language

- Science Perspective
 - Vision is how we observe and understand the world
 - Language is how we communicate
- A move towards Artificial General Intelligence (AGI)



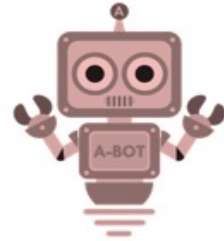
Applications at the intersection of Vision and Language

- To aid “visually impaired” people



Applications at the intersection of Vision and Language

- To aid “situationally impaired” analysts



Did anyone enter this room last week?

Yes, 127 instances logged on camera

Show me images of anyone carrying a black bag.

...



Applications at the intersection of Vision and Language

➤ Personal Assistants



And what's next?

Add the chopped zucchini to the pan...

Hey, can you order more zucchini?

Done.



Applications at the intersection of Vision and Language

➤ Natural Language Instructions for Robots



Is there smoke in any room around you?

Yes, in one room

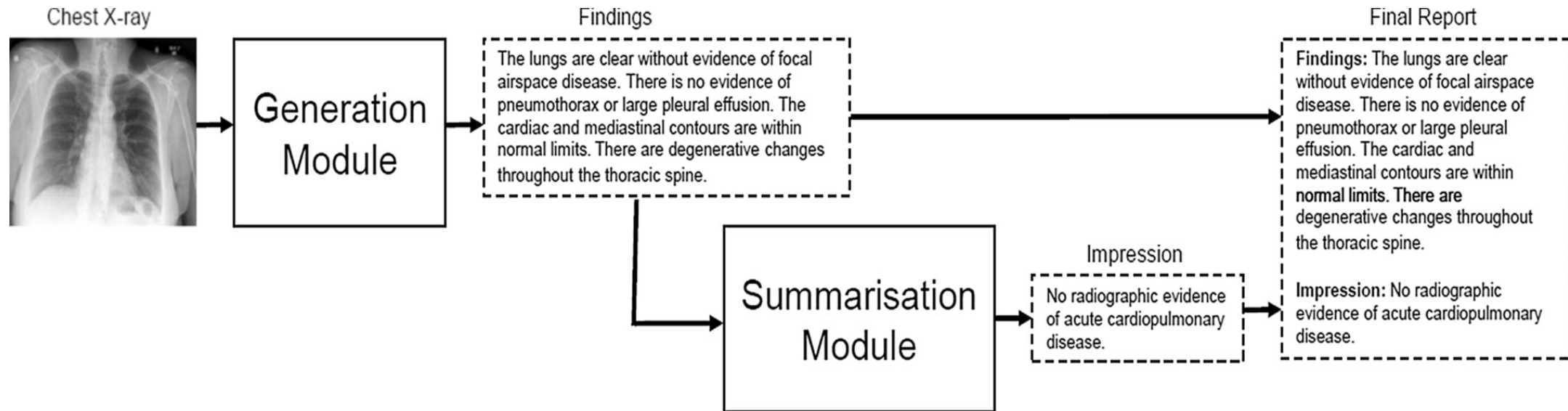
Go there and look for people

...



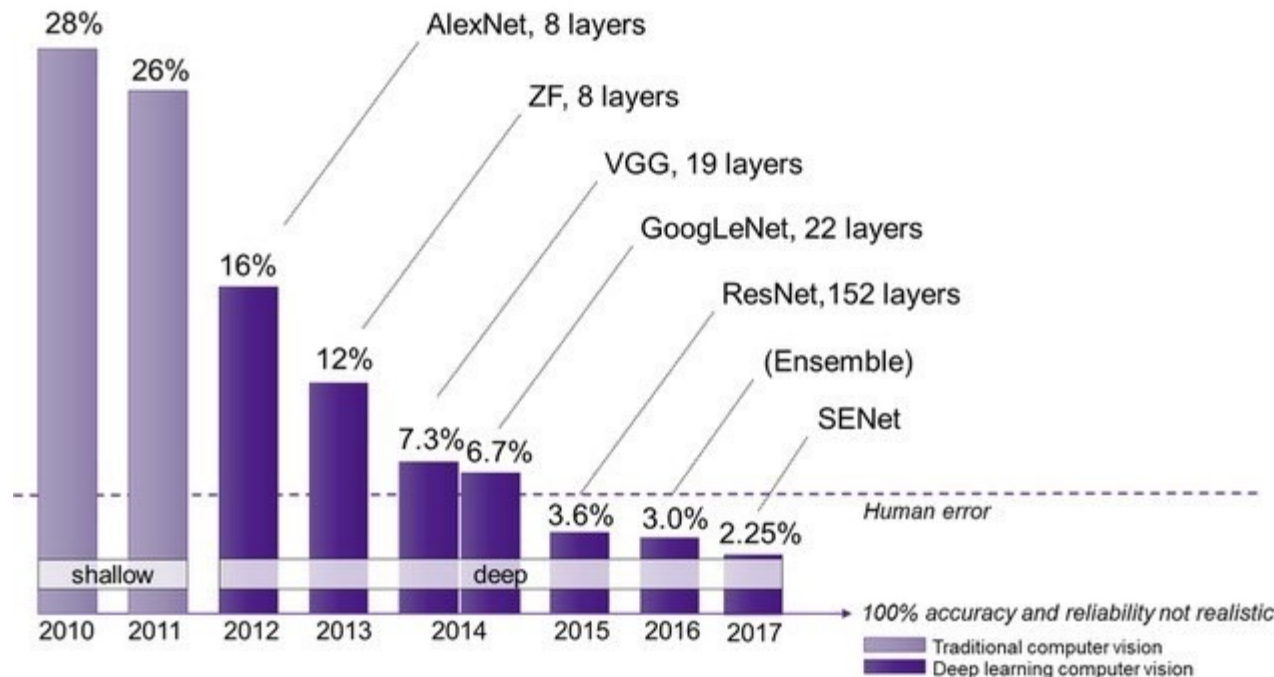
Applications at the intersection of Vision and Language

- Generating and summarizing radiology reports from medical images



Building Blocks: Convolutional Neural Networks (CNNs)

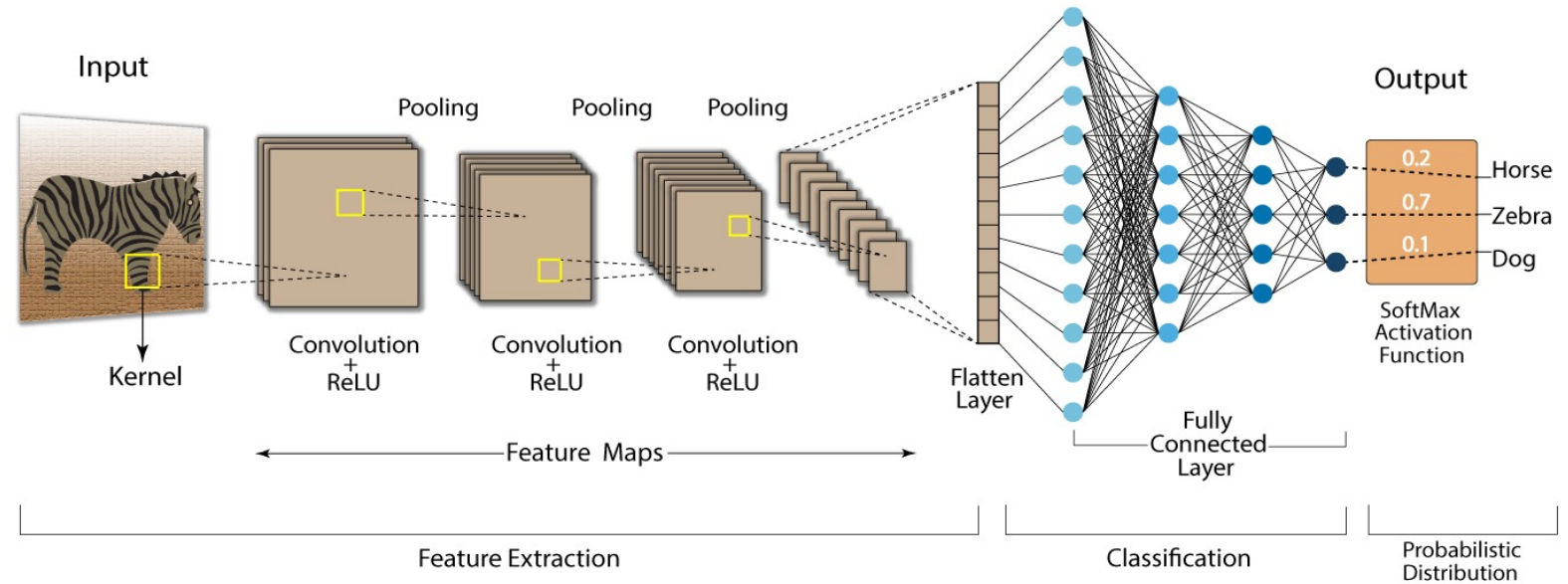
- A class of deep neural networks suitable for processing 2D/3D data. For e.g., Images and Videos
- CNNs can capture high-level representation of images/videos which can be used for end-tasks such as classification, object detection, segmentation, etc.
- A range of CNNs improving over the years



CNN Architecture

➤ A typical CNN architecture consists of the following layers:

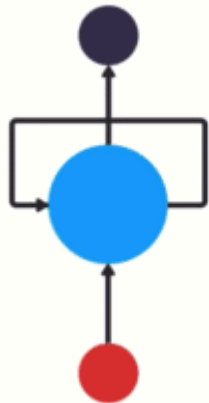
- Convolution layer
- ReLU layer (non-linearity)
- Pooling layer
- Flattening
- Fully-connected layer
- Output layer



➤ There can be multiple steps of convolution followed by pooling, before reaching the fully connected layers.

Building Blocks: Recurrent Neural Networks (RNNs)

- A class of neural networks suitable for processing temporal or sequential data
- The basic unit of RNN is called “cell”, and each cell consists of layers and a series of cells that enables the sequential processing of recurrent neural network models
- RNNs have a looping mechanism that acts as a highway to allow information to flow from one step to the next. This information is the hidden state, which is a representation of previous inputs.



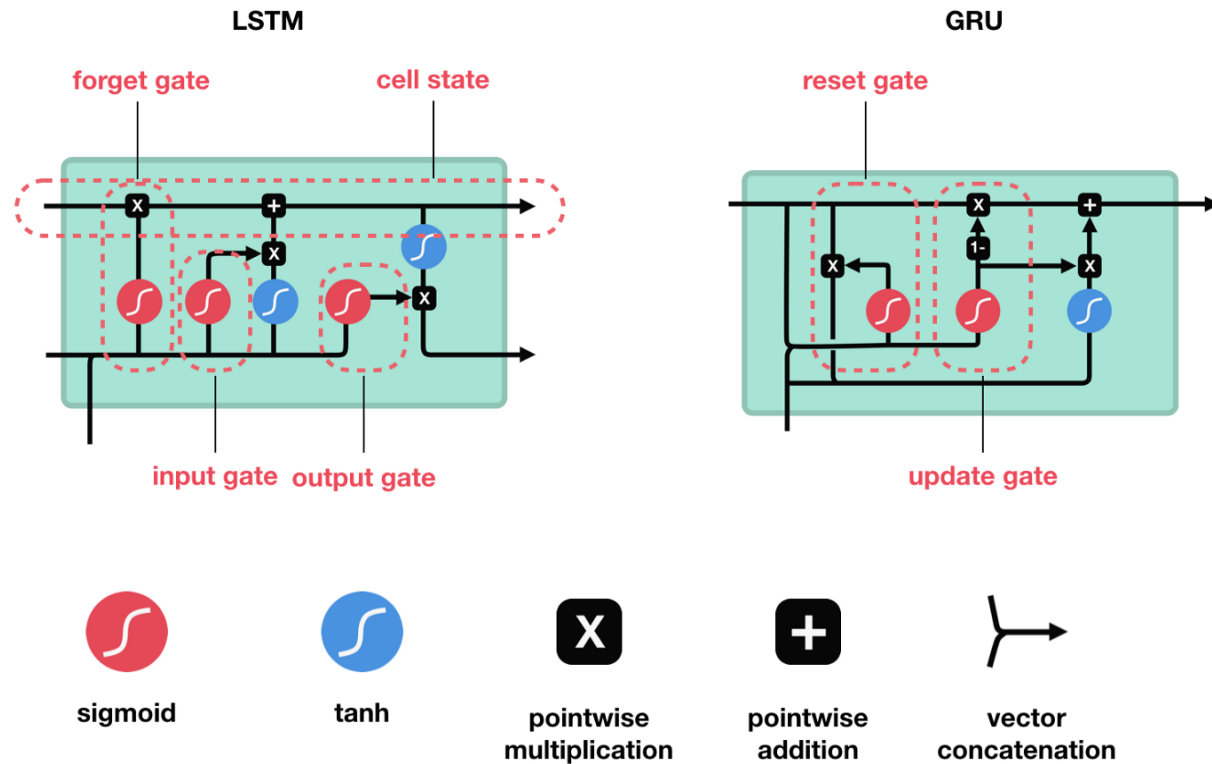
Long-range Dependency problem

- Vanilla RNNs suffers from vanishing gradient problem
 - As the RNNs processes more steps, it has troubles retaining information from previous steps.
 - Due to back-propagation, the earlier layers fail to do any learning as the internal weights are barely being adjusted due to extremely small gradients.
 - Does not learn the long-range dependencies across time steps

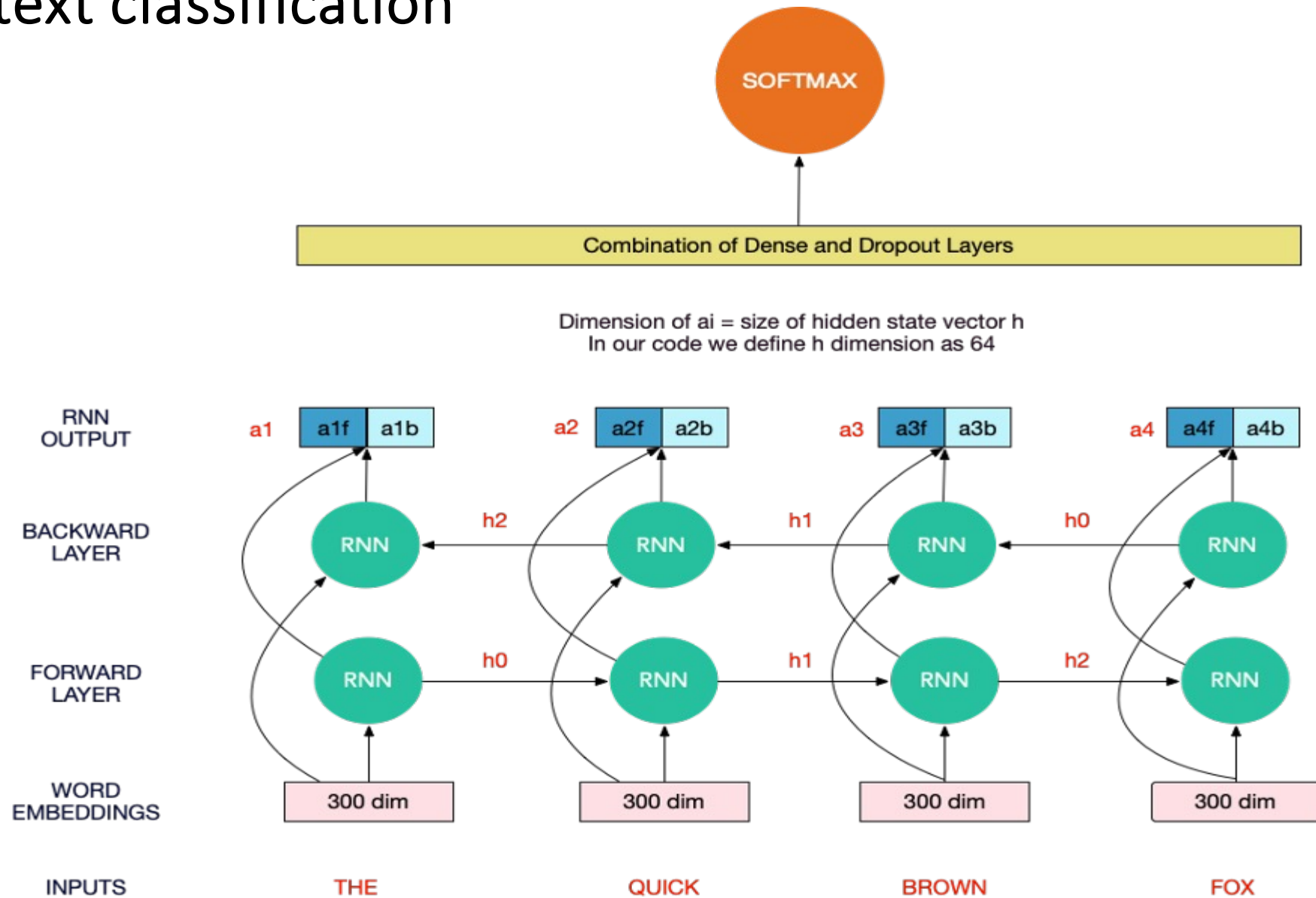
“Once upon a time, there was a king who ruled a great and glorious nation. Favourite amongst his subjects was the court painter of whom he was very proud. Everybody agreed this wizzened old man painted the greatest pictures in the whole kingdom and the king would spend hours each day gazing at them in wonder. However, one day a dirty and disheveled stranger presented himself at the court claiming that in fact he was the greatest painter in the land. The indignant king decreed a competition would be held between the two artists, confident it would teach the vagabond an embarrassing lesson. Within a month they were both to produce a masterpiece that would out do the other. After thirty days of working feverishly day and night, both artists were ready. They placed their paintings, each hidden by a cloth, on easels in the great hall of the castle. As a large crowd gathered, the king ordered the cloth be pulled...”

LSTMs/GRUs

- LSTMs and GRUs are two special RNNs, capable of learning long-term dependencies using mechanisms called **gates**.
- These gates are different tensor operations that can learn what information to add or remove to the hidden state.

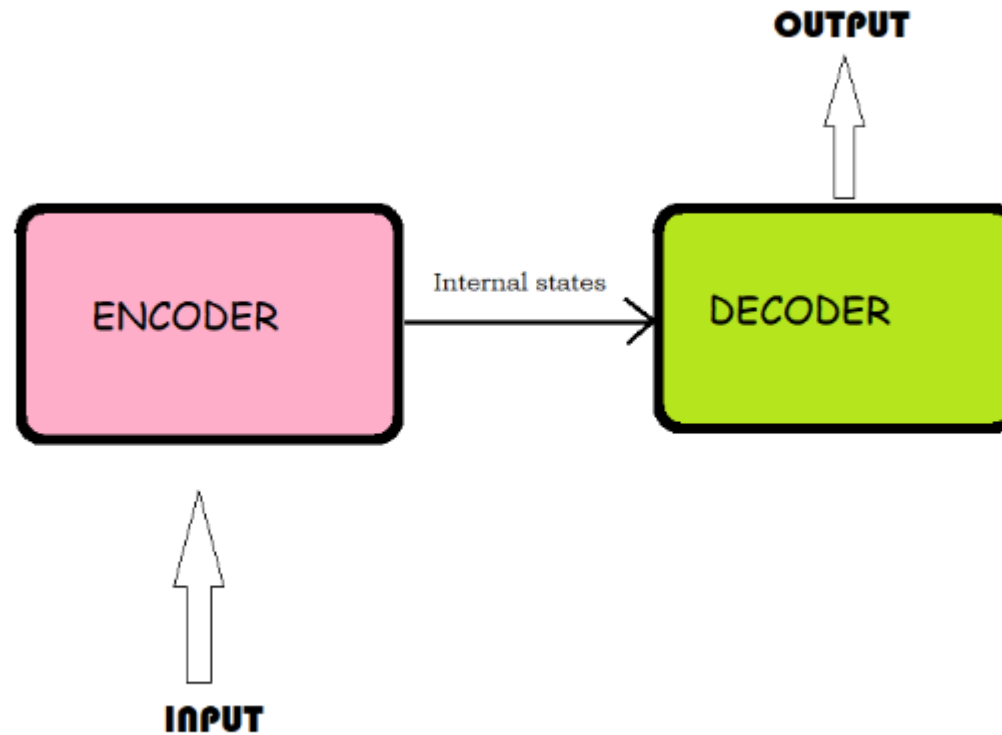


End-to-End text classification



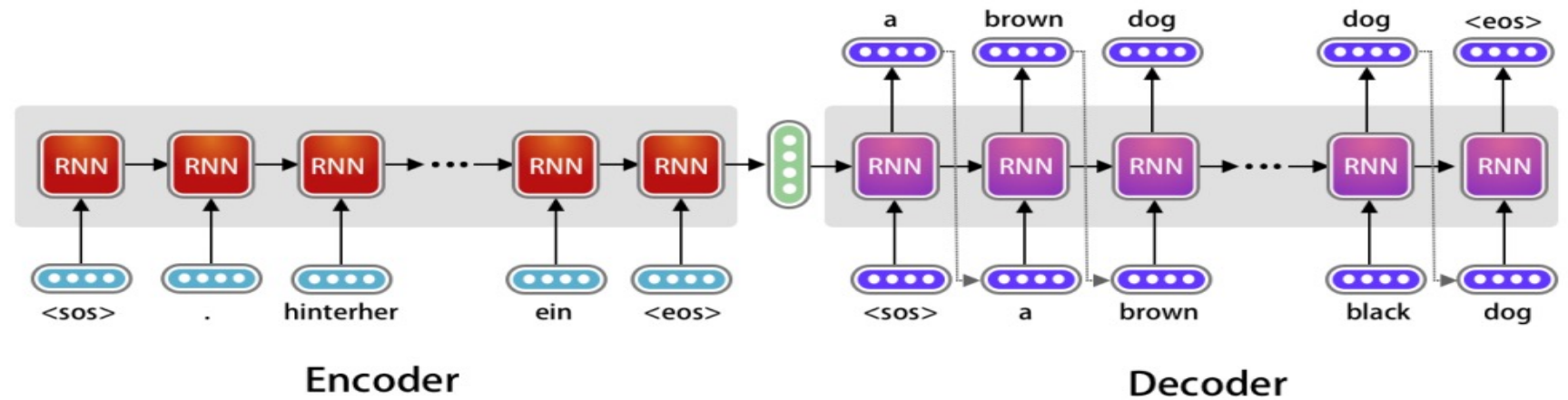
Seq2Seq model

- Model has two parts: Encoder and Decoder (Encoder-Decoder Framework)

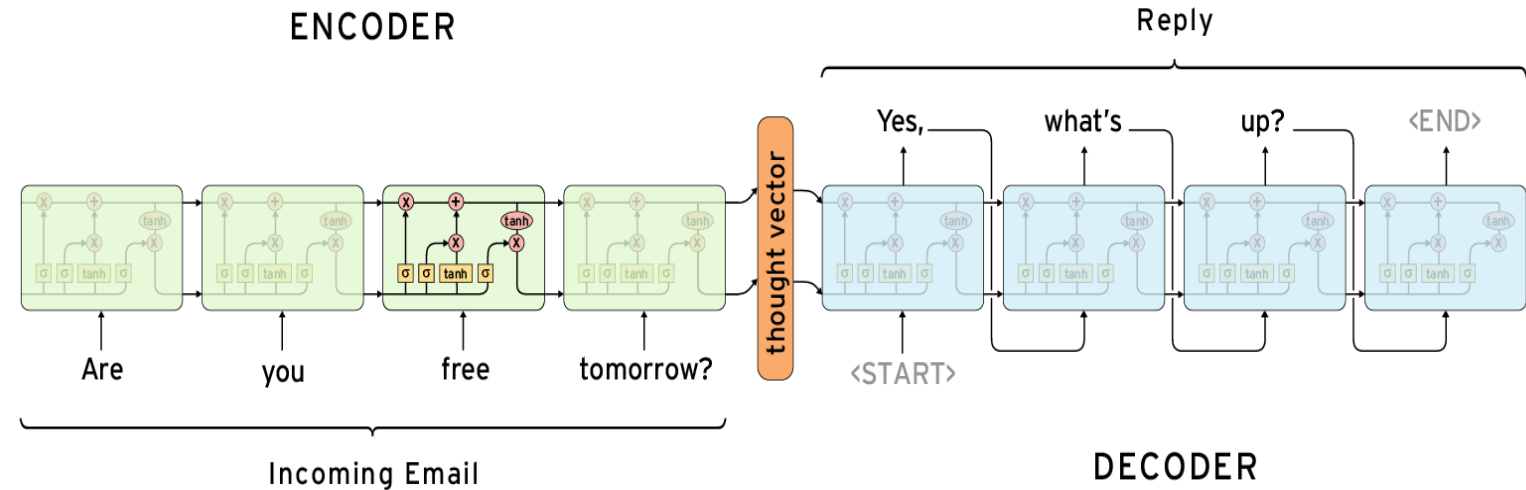


Seq2Seq used for various NLP applications

➤ Machine Translation

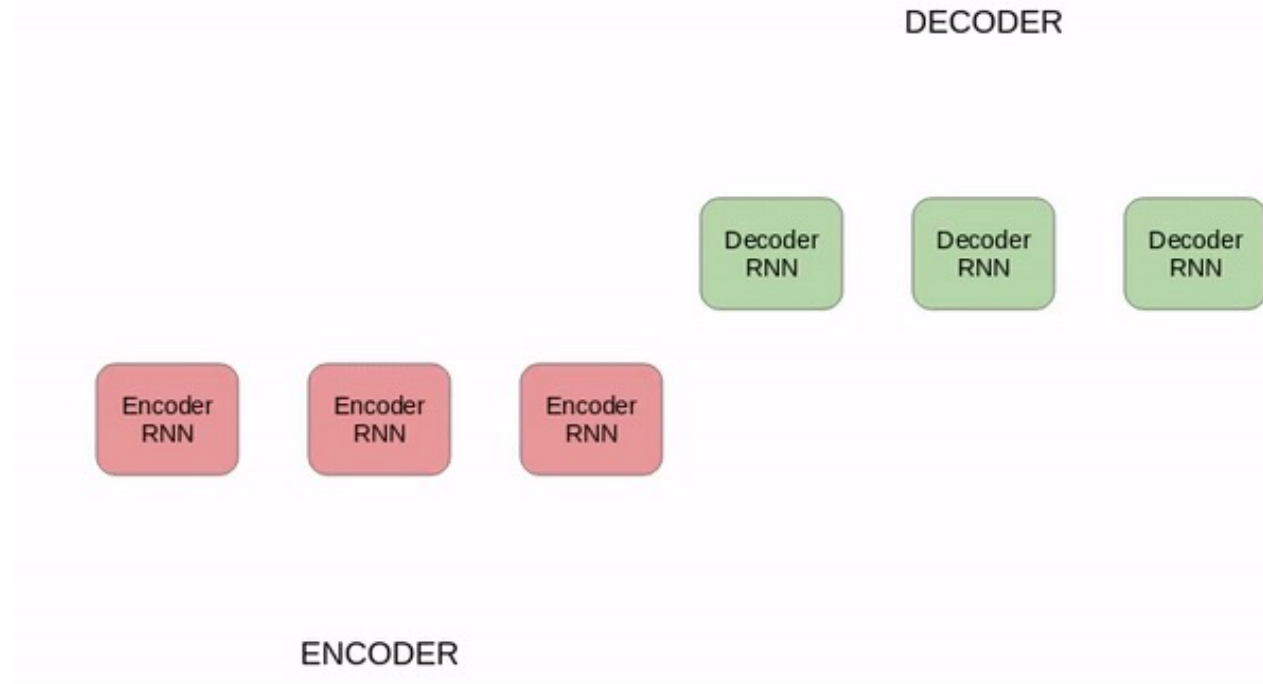


➤ Automatic Email Reply



Issues with RNNs for seq2seq tasks

- Despite being very successful for various NLP tasks, there were challenges:
 - dealing with long-range dependencies
 - the sequential nature of the architecture prevents parallelization
- Attention mechanism helped to overcome first issue to certain extent



Attention Mechanism

- A set of mechanisms that limit some processing to a subset of incoming stimuli (reducing computational demands)
- Attention in neural networks
 - A mechanism that **learns to focus** on a subset of the **input** that is **relevant to the task**.

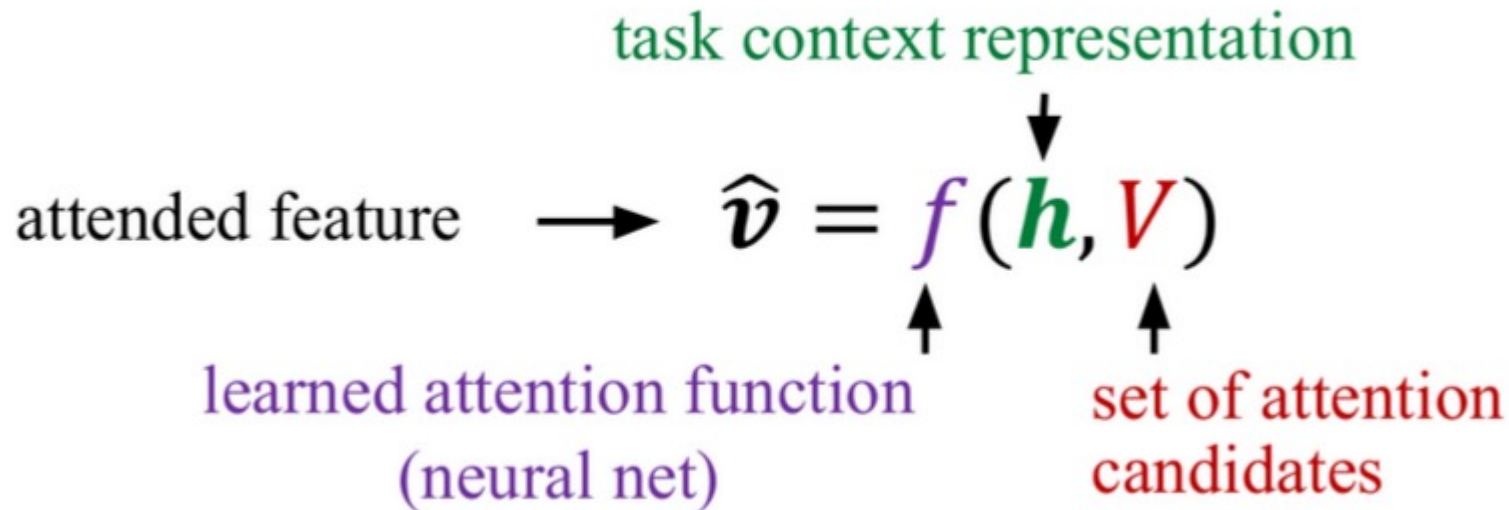
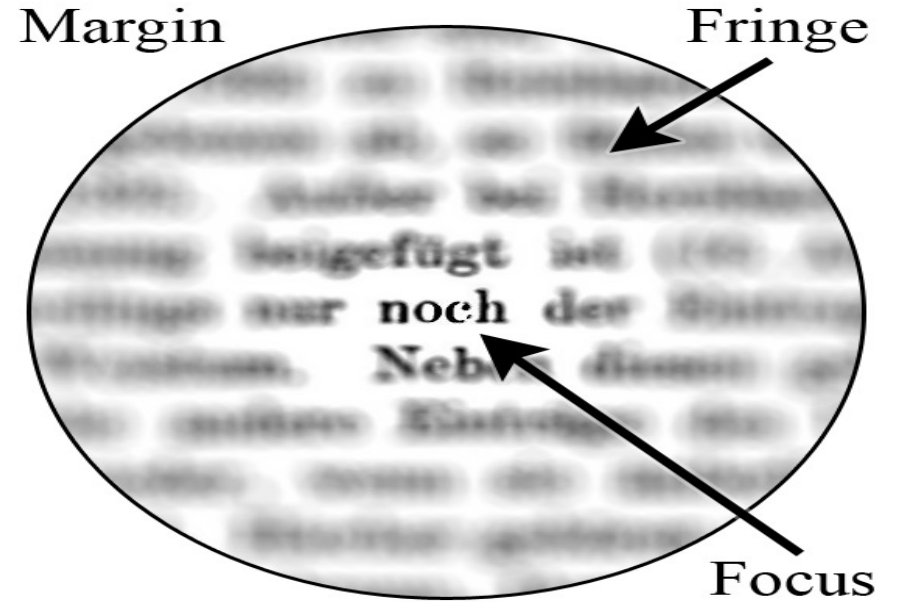


Image Captioning

- Developing models to generate textual description of an image



"man in black shirt is playing guitar."

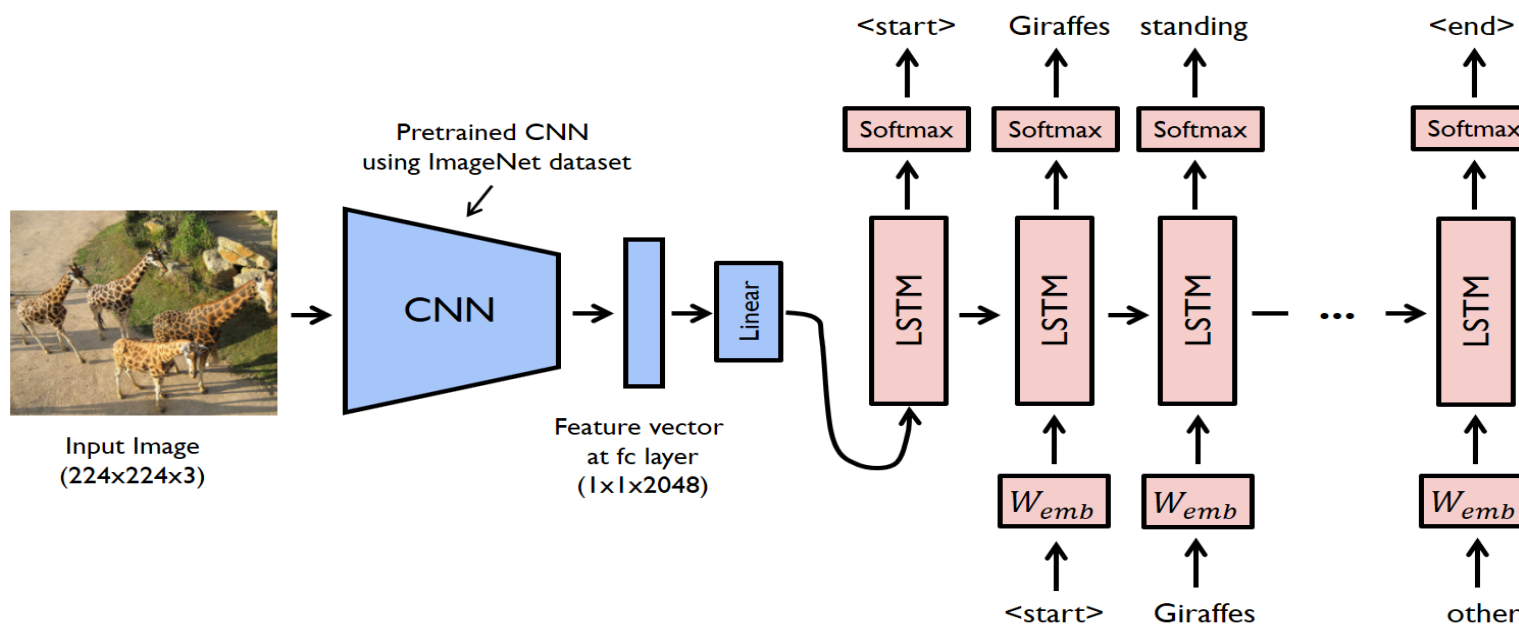
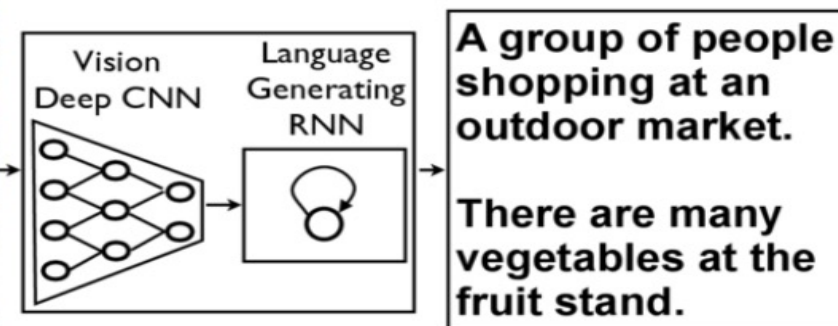


"construction worker in orange safety vest is working on road."



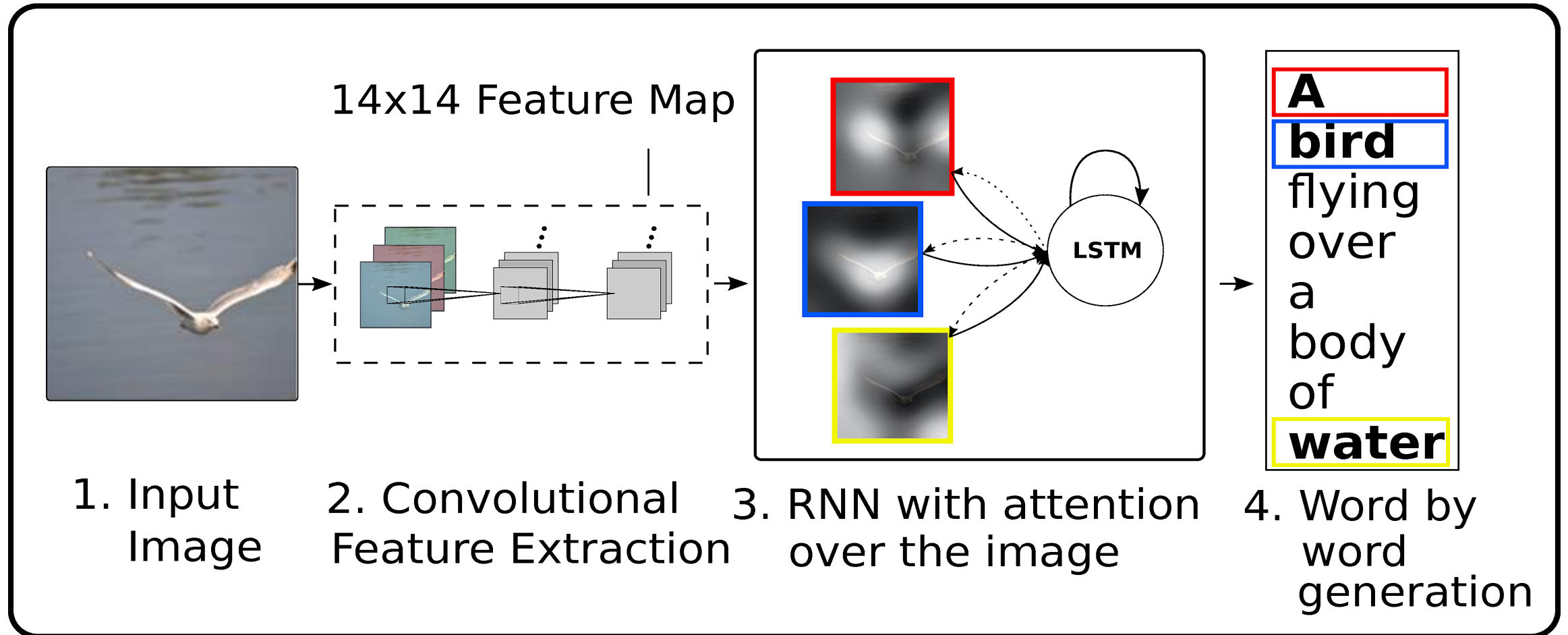
"two young girls are playing with lego toy."

Show and Tell (Vinyals et al., 2015)



Show, Attend and Tell (Xu et al., 2015)

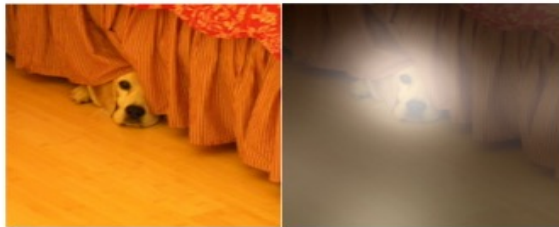
- “Rather than compressing an entire image into static representation, attention allows for **salient features** to dynamically come to forefront as needed”



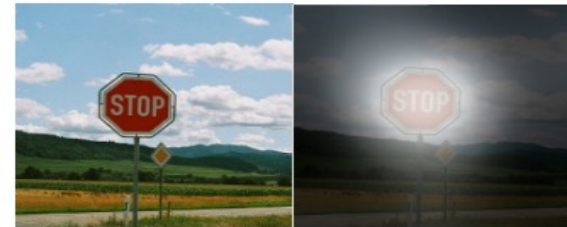
Show, Attend and Tell (Xu et al., 2015)



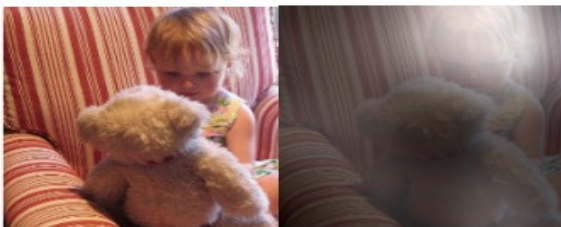
A woman is throwing a frisbee in a park.



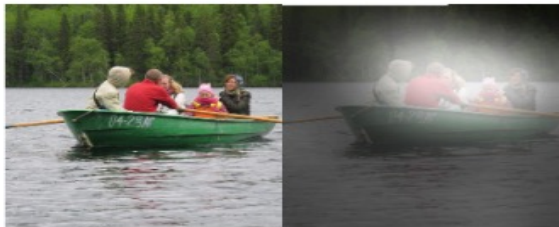
A dog is standing on a hardwood floor.



A stop sign is on a road with a mountain in the background.



A little girl sitting on a bed with a teddy bear.



A group of people sitting on a boat in the water.



A giraffe standing in a forest with trees in the background.

<start>



a



large



airplane



flying



in



the



blue



sky



<end>

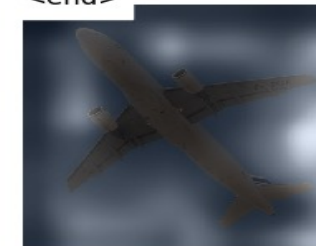


Image Captioning with Semantic Attention (You et al., 2016)

- Learns to selectively attend to semantic concept proposals and fuse them into hidden states and outputs of RNNs.

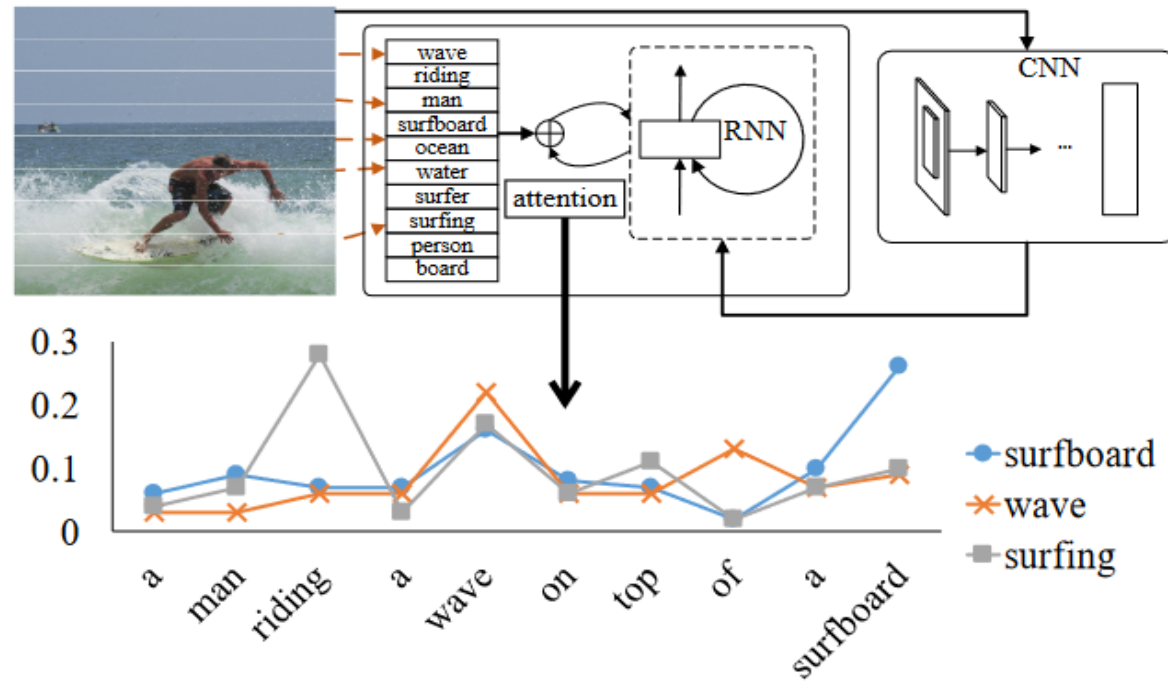
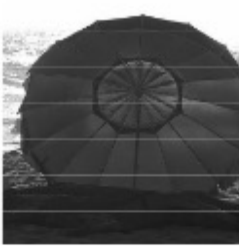


Image Captioning with Semantic Attention (You et al., 2016)

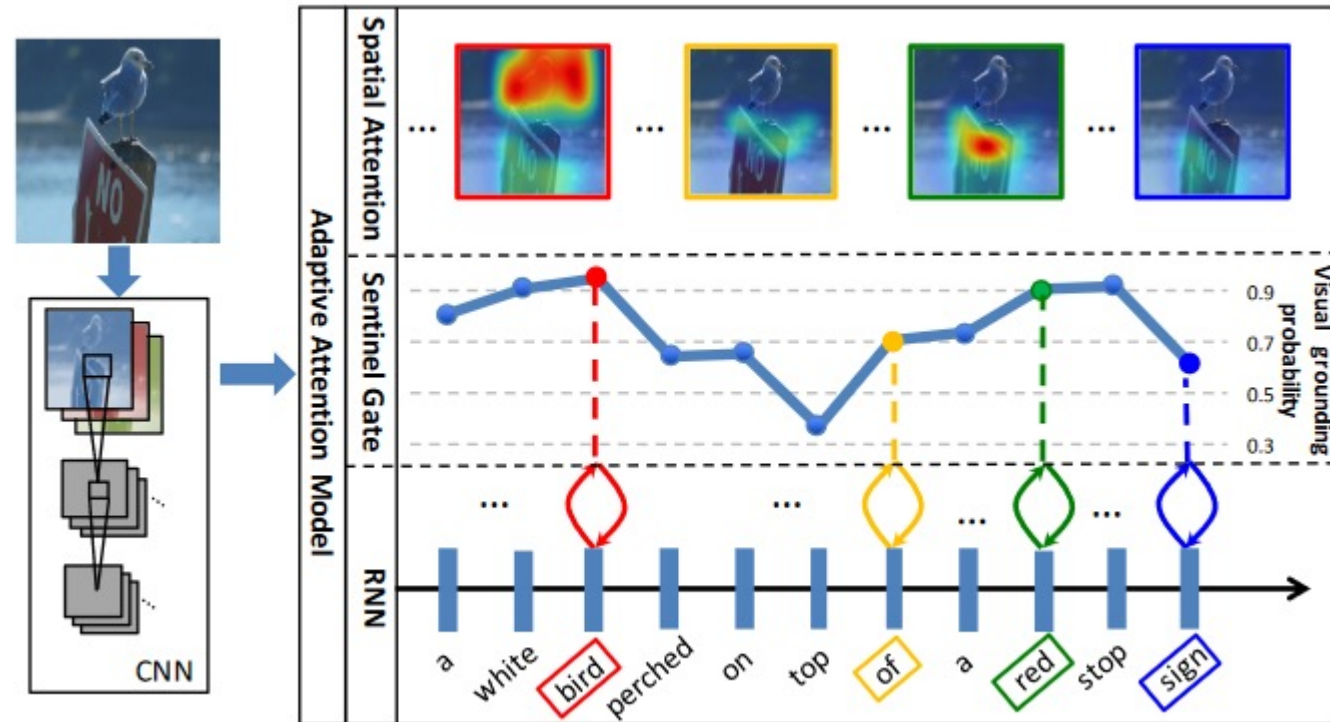
- Learns to selectively attend to semantic concept proposals and fuse them into hidden states and outputs of RNNs.

						
Google NIC	a white plate topped with a variety of food.	a baby is eating a piece of paper.	a close up of a plate of food on a table.	a teddy bear sitting on top of a chair .	a person is holding colorful umbrella.	a woman is holding a cell phone in her hand .
Top-5 visual attributes	plate broccoli fries food french	teeth brushing toothbrush holding baby	cake table plate sitting birthday	teddy cat bear stuffed white	umbrella beach water sitting boat	woman bathroom her scissors man
ATT-FCN	a plate with a sandwich and french fries.	a baby with a toothbrush in its mouth.	a table topped with a cake with candles on it.	a white teddy bear sitting next to a stuffed animal	a black umbrella sitting on top of a sandy beach .	a woman holding a pair of scissors in her hands .

	
a traffic light is on a city street.	a yellow and black train on a track.
street sign cars clock traffic	train tracks clock tower down
a street with cars and a clock tower.	a train traveling down tracks next to a building.

Knowing When to Look (Lu et al., 2017)

- Words such as “a”, “of”, “it” may be seen as not worth attending
- Words such as “woman”, “dog”, “traffic light” need attending to the image
- Automatically determines when to look (sentinel gate) and where to look (spatial attention) for word generation.



Knowing When to Look (Lu et al., 2017)

- Visualization of generated captions and image attention maps on COCO dataset



a little girl sitting on a bench holding an umbrella.



a herd of sheep grazing on a lush green hillside.



a close up of a fire hydrant on a sidewalk.



a yellow plate topped with meat and broccoli.



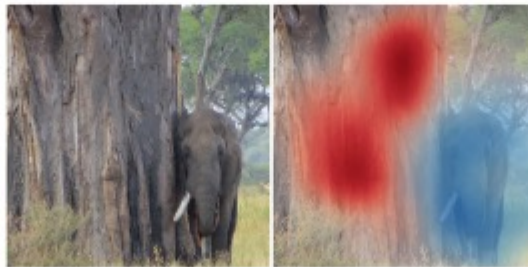
a zebra standing next to a zebra in a dirt field.



a stainless steel oven in a kitchen with wood cabinets.



two birds sitting on top of a tree branch.



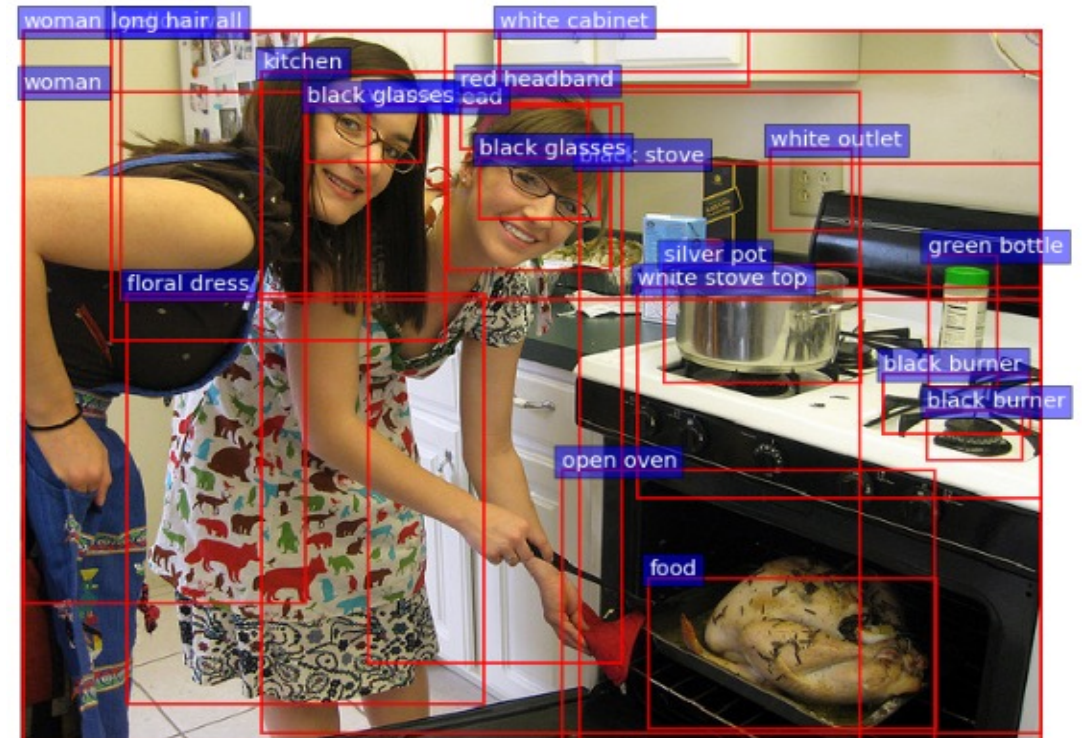
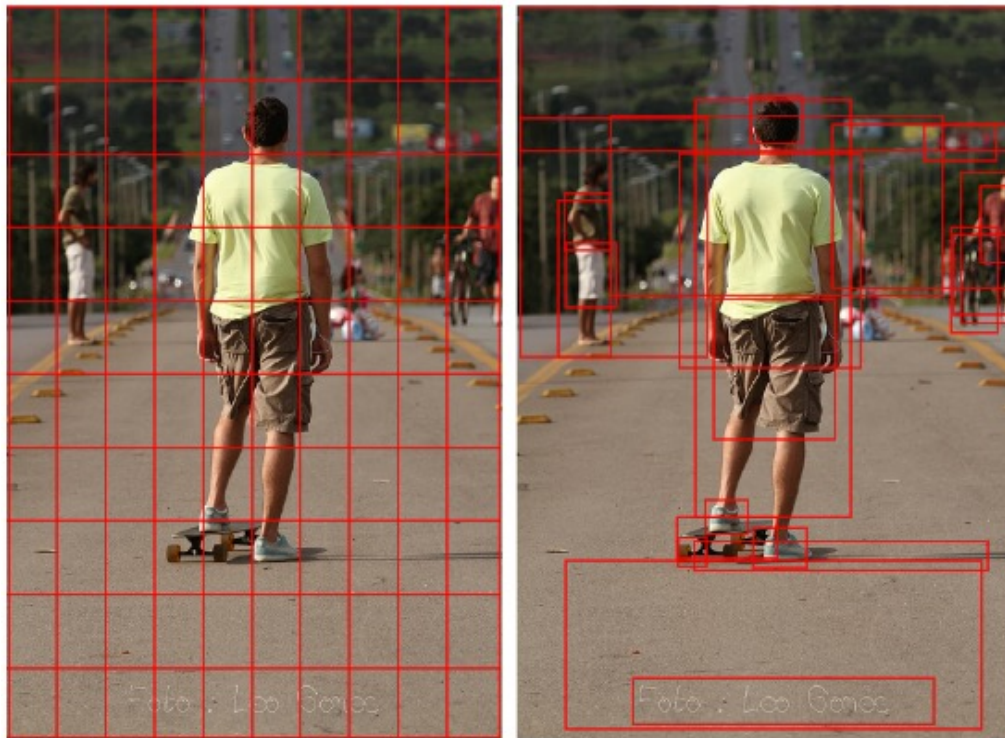
an elephant standing next to rock wall.



a man riding a bike down a road next to a body of water.

Bottom-up and Top-down attention (Anderson et al., 2018)

- Previous attention models operate on CNN features corresponding to a uniform grid of equally-sized image regions
- Bottom-up and Top-down enables attention to be calculated at the level of objects (or salient image regions)
- Features vectors extracted from Faster R-CNN are used



Bottom-up and Top-down attention (Anderson et al., 2018)



Ours: Resnet – A man sitting on a *toilet* in a bathroom.



Ours: Up-Down – A man sitting on a *couch* in a bathroom.

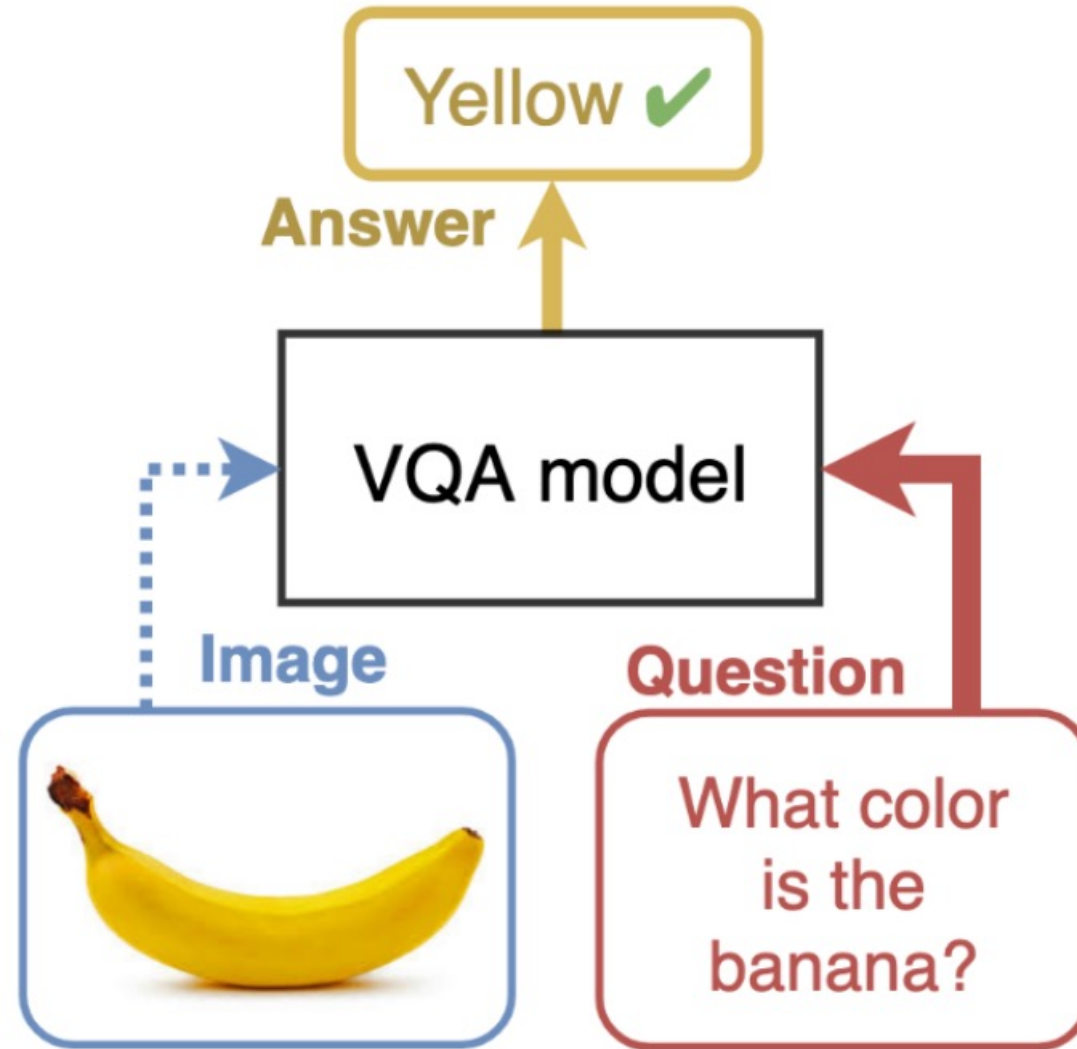


And many more work ...

- Show, Control and Tell: A Framework for Generating Controllable and Grounded Captions – Cornia M et al., CVPR 2019
 - Pointing Novel Objects in Image Captioning, Li Y et al., CVPR 2019
 - Describing like humans: on diversity in image captioning – Wang Q et al., CVPR 2019
 - Length-Controllable Image Captioning – Deng C et al., ECCV 2020
 - Transformer-based local-global guidance for image captioning – Parvin et al., 2023
 - Show, tell and summarise: learning to generate and summarise radiology findings from medical images – Singh et al., Neural Computing & Applications, 2021.
 - Medical image captioning via generative pretrained transformers – Selivanov et al. Scientific Reports, 2023
-
- Let's see Image captioning in action
 - <https://milhidaka.github.io/chainer-image-caption/>

Visual Question Answering (VQA)

- Given an **image**, can our machine answer the corresponding **questions in natural language**?



VQA Dataset



Who is wearing glasses?

man



woman

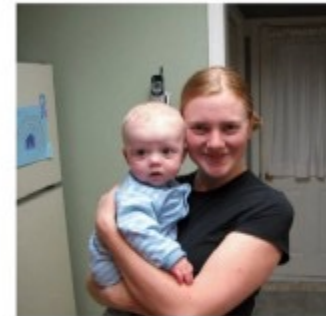


Where is the child sitting?

fridge



arms



Is the umbrella upside down?

yes



no



How many children are in the bed?

2

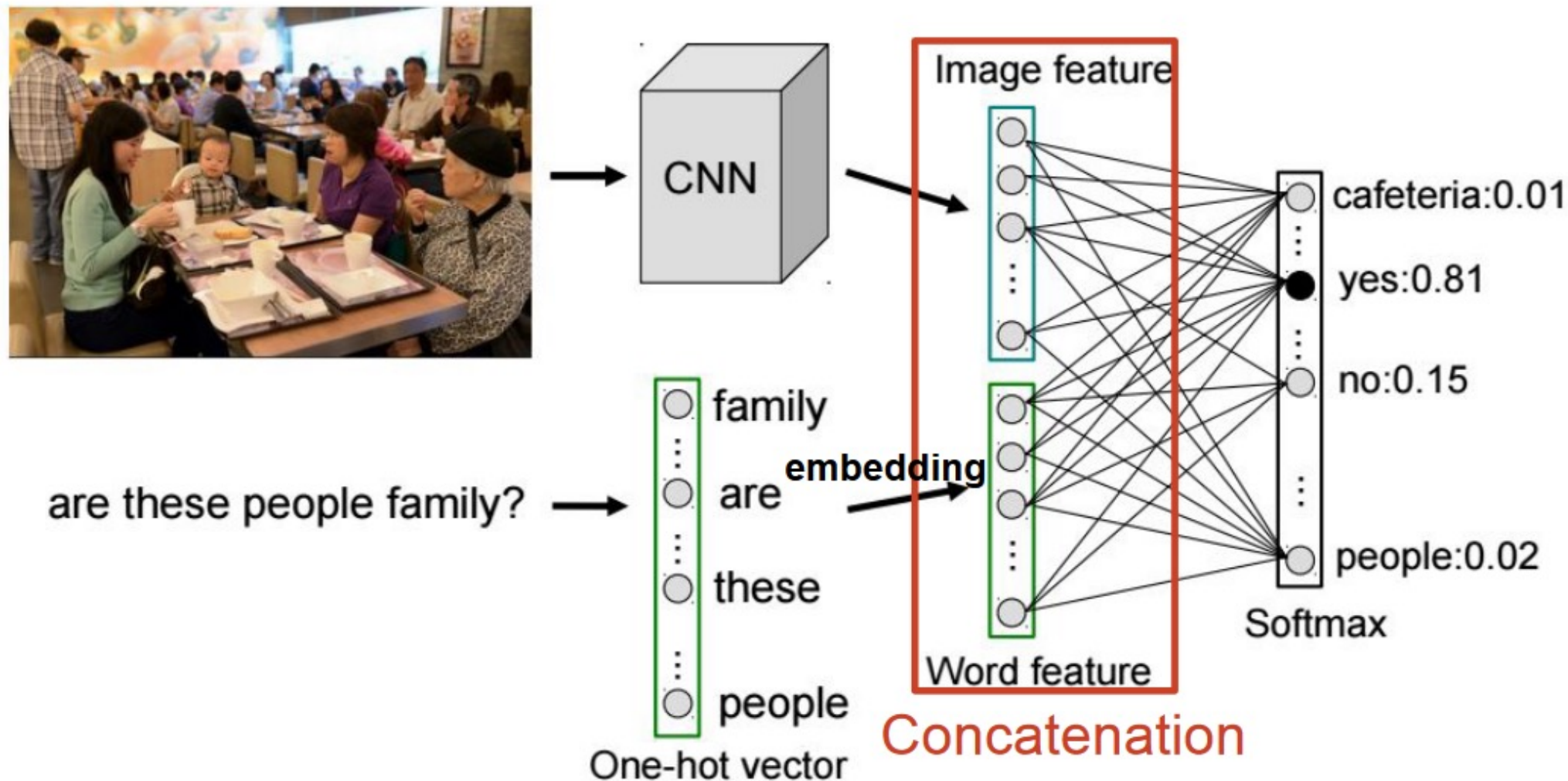


1



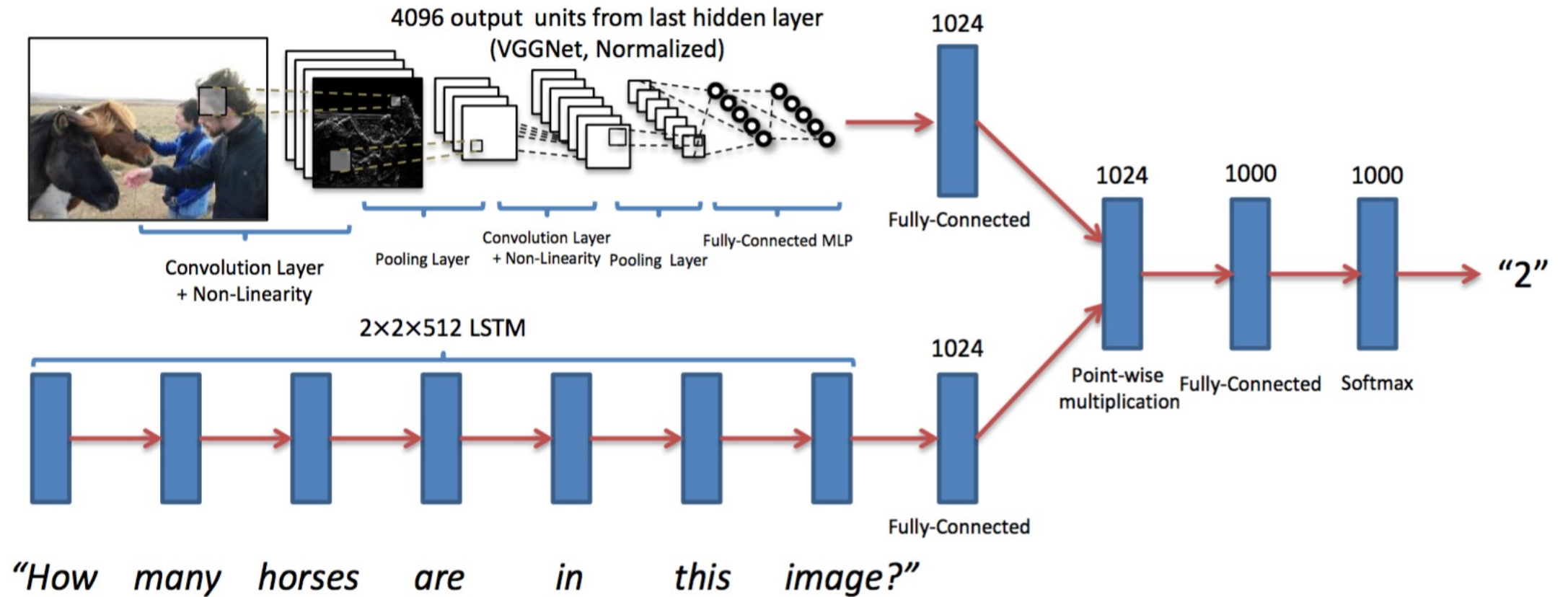
VQA Approach: Bag-of-words + Image feature (iBOWIMG)

- Combine image and word embeddings to predict answer

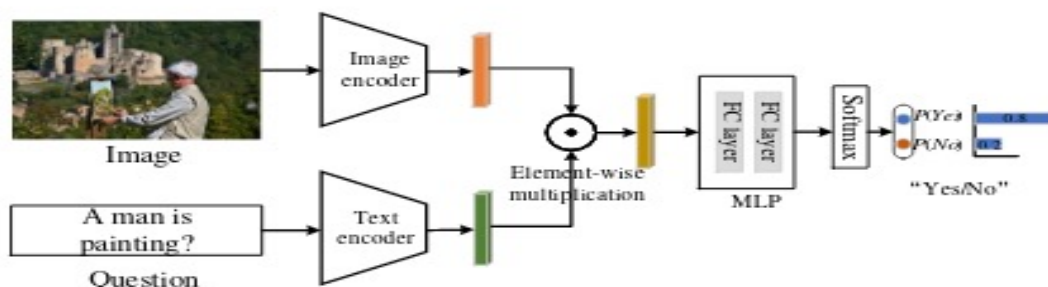


VQA Common Approach

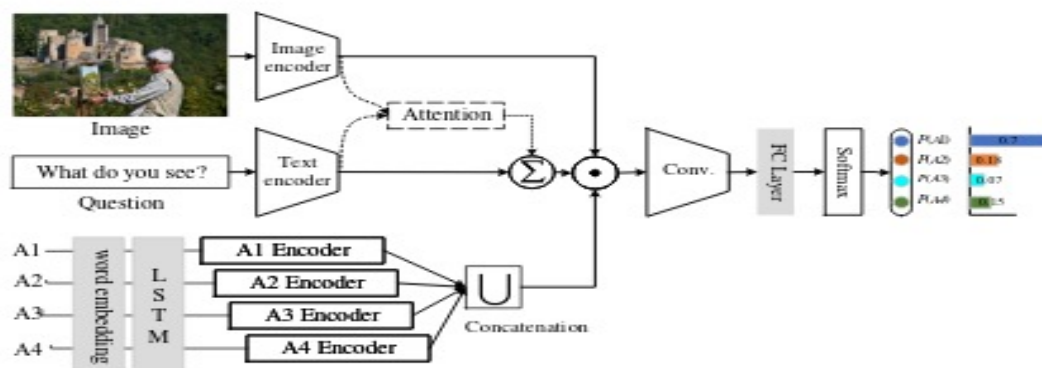
- Combine CNN (for vision) and RNN (for language) to predict answer



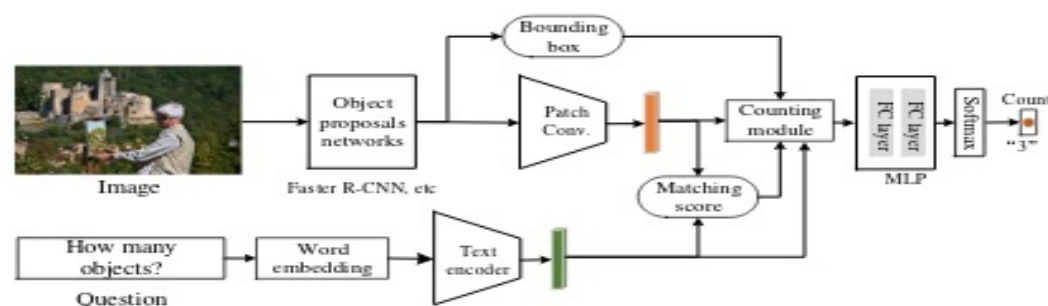
VQA – set of problems



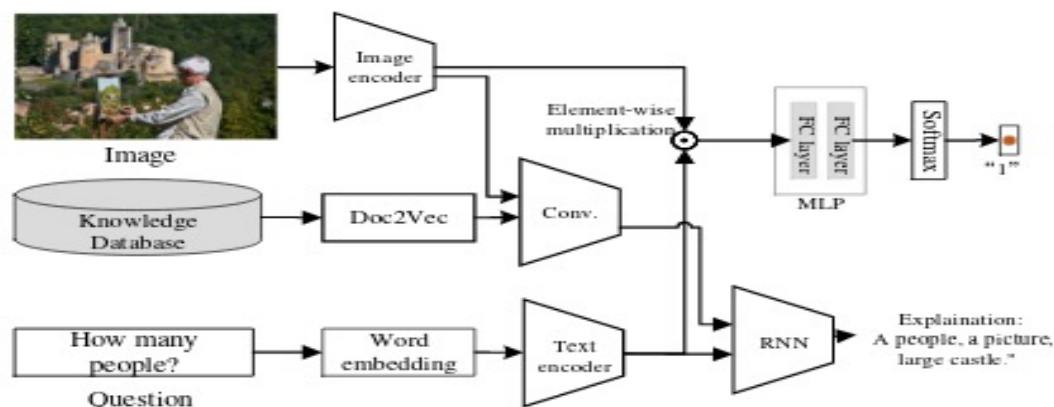
(a) “Yes/no” problem



(b) Multi-choice problem



(c) Number counting problem

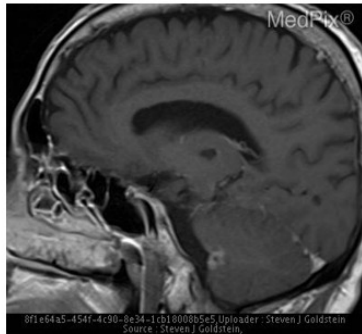


(d) Open-ended problem

Figure 2 Common types of visual question answering. “Yes/No” problem and multi-choice problem can be regarded as a classification problem, while number counting problem and open-ended problem can be viewed as a caption generation problem.

VQA in the Medical Domain

- Aims to create a system that can answer natural language questions based on a given medical image.
- VQA-Med 2019 Challenge dataset



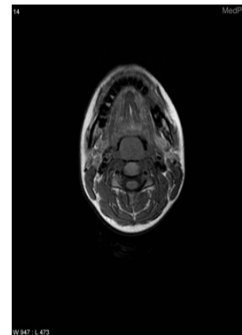
Q: What is the plane of this MRI?
A: Sagittal



Q: The CT scan shows what organ system?
A: Spine and contents

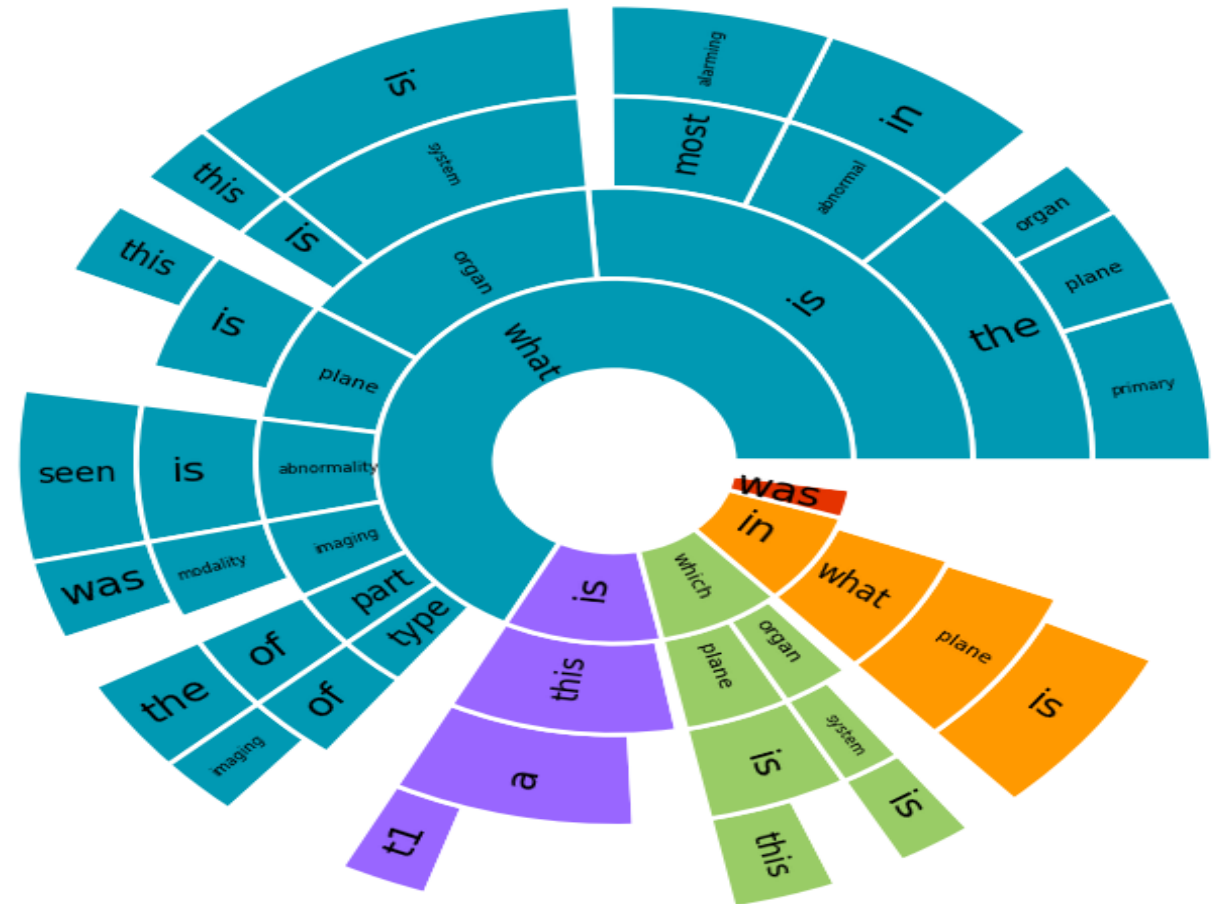


Q: With what modality is this image taken?
A: AN - angiogram



Q: What is most alarming about this MRI?
A: Schwannoma

- Best strategy is to do classification rather than generation



VQA-Med Methodology

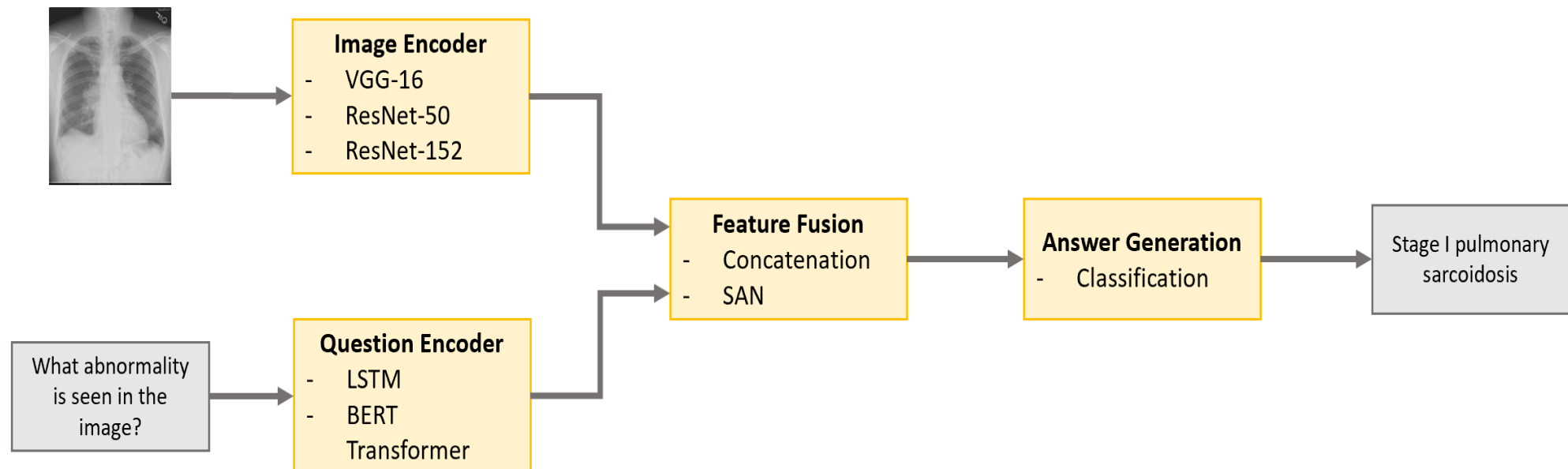
➤ Contributions

➤ Incorporating medical domain knowledge

- Image encoder – applied self-supervised pretraining using Radiology Objects in COntext (ROCO) dataset
- Question encoder – used BioBERT, pretrained on same tasks as BERT, but using the PubMed corpus

➤ Evidence verification

- Gradient Weighted Class Activation Map (Grad-CAM) for evidence verification



Results

- Test accuracy achieved by each model variation

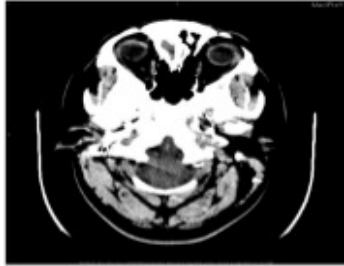
Model Variation	Test Accuracy
VGG-16 + LSTM + Concatenation	0.56
ResNet-50 + LSTM + Concatenation	0.54
ResNet-152 + LSTM + Concatenation	0.53
VGG-16 + BERT + Concatenation	0.60
VGG-16 + BERT + SAN	0.58
VGG-16 + BioBERT + Concatenation	0.60
Pretrained VGG-16 + BERT + Concatenation	0.60

- Accuracy of the baseline vs. BERT model per category type

Model Variation	Baseline	+BERT
Modality	0.64	0.76
Plane	0.78	0.77
Organ	0.74	0.74
Abnormality	0.06	0.08
Overall	0.56	0.60

Results

➤ Effect of using BERT vs. LSTM



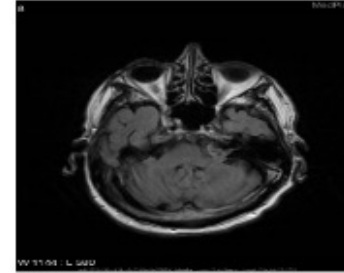
Q: What imaging modality was used to take this image?
GT: CT with IV contrast
Baseline: Skull fracture from cell phone
+BERT: CT with IV contrast

(a) Category misclassification



Q: What was this image taken with?
GT: MR – PDW proton density
Baseline: Yes
+BERT: MR – PDW proton density

(b) Wrong answer type

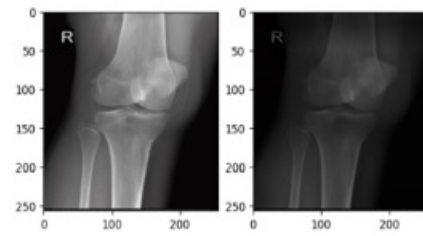


Q: Is this a contrast or noncontrast MRI?
GT: Noncontrast
Baseline: MR – flair
+BERT: Noncontrast

(c) Not choosing from options

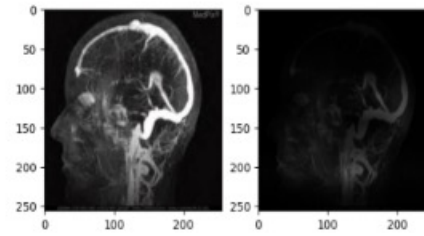
Evidence verification

- Attention distribution output by Stacked Attention Network (SAN) fusion method



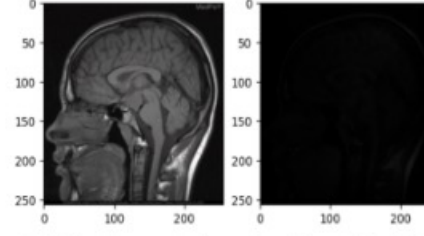
Q: What imaging modality is seen here?
GT: XR – plain film
A: XR – plain film

(a)



Q: What plane is demonstrated?
GT: Axial
A: Coronal

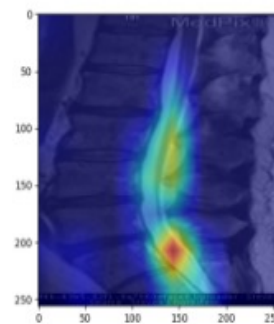
(b)



Q: What is most alarming about this MRI?
GT: Hypothalamic hamartoma
A: Meningioma

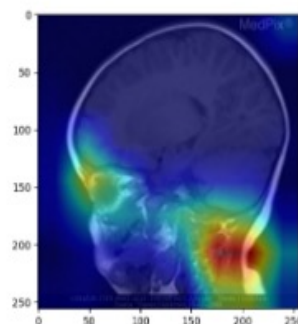
(c)

- Grad-CAM output



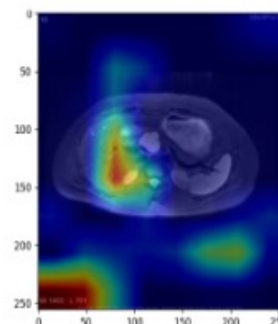
Q: Is this a T1 weighted image?
GT: No
A: No

(a)



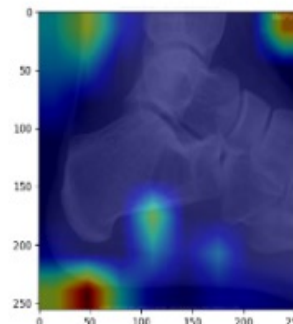
Q: In what plane is this MRI captured?
GT: Sagittal
A: Sagittal

(b)



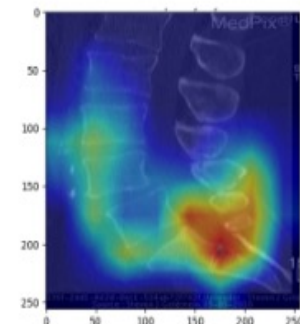
Q: What part of the body is being imaged?
GT: Gastrointestinal
A: Gastrointestinal

(c)



Q: What is abnormal in the X-ray?
GT: Chondroblastoma
A: Jones fracture and dancer's fracture

(d)



Q: What abnormality is seen in the image?
GT: Spondylolisthesis bilateral pars fracture
A: Spondylolysis

(e)

Let's see in action

- <https://visualqa.org/visualize/>
- <https://demo.allennlp.org/visual-question-answering>

Visual Dialog

- Ability to hold a meaningful dialog with humans in natural language about visual content
- **Given an image I, a history of dialog consisting of sequence of question-answer pairs, and a natural language follow-up question, the task for the machine is to answer the question in free-form natural language.**



Image Captioning vs. VQA vs. Visual Dialog



VQA

Q: How many people
on wheelchairs ?

A: Two

Q: How many wheelchairs ?

A: One

Captioning

Two people are in a
wheelchair and one is
holding a racket.

Visual Dialog

Q: How many people are on
wheelchairs ?

A: Two

Q: What are their genders ?

A: One male and one female

Q: Which one is holding a
racket ?

A: The woman



Visual Dialog

Q: What is the gender of the
one in the white shirt ?

A: She is a woman

Q: What is she doing ?

A: Playing a Wii game

Q: Is that a man to her right ?

A: No, it's a woman

Session
Variables

Visual Dialog – Late Fusion Encoder

- Entire history H is concatenated and encoded by LSTM



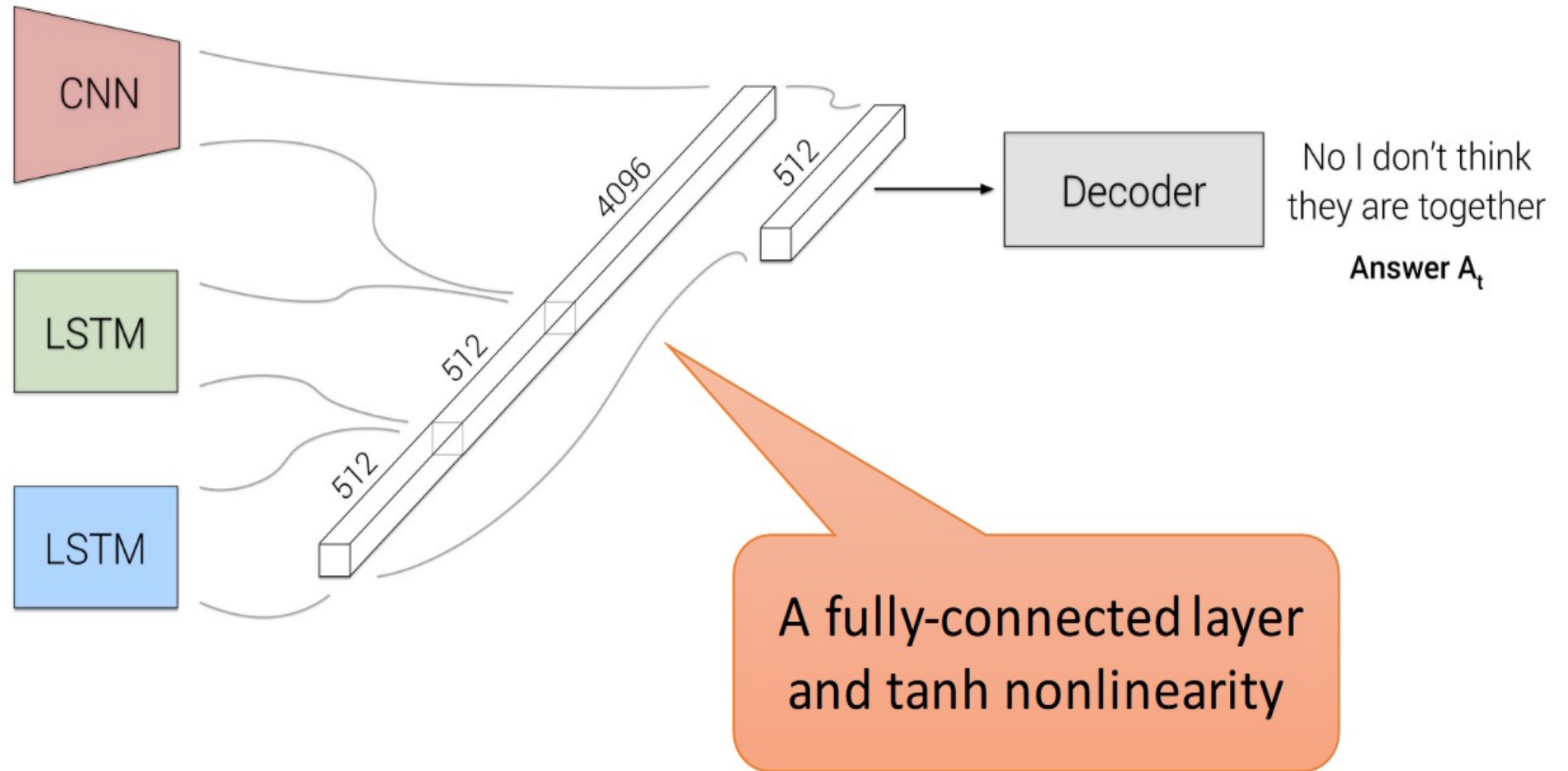
Image I

Do you think the woman is with him?

Question Q_t

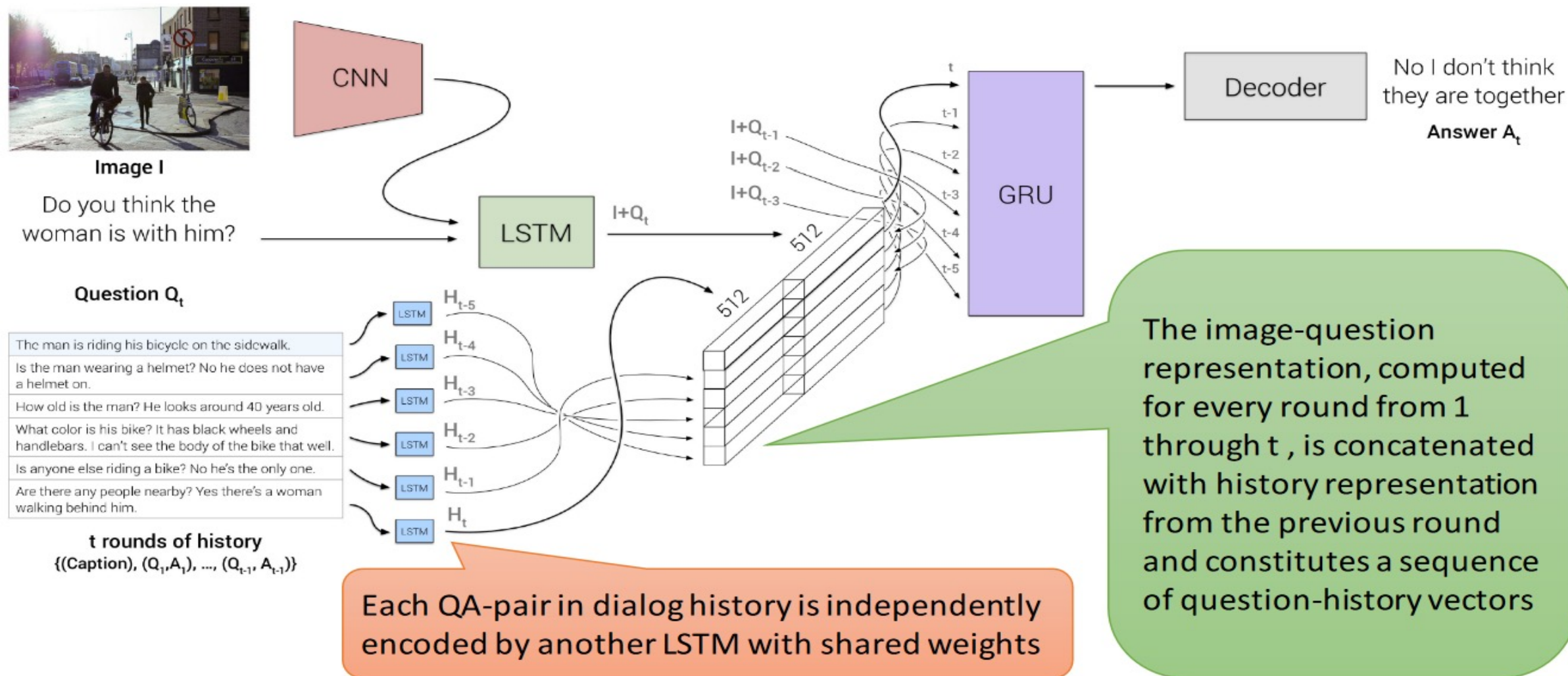
The man is riding his bicycle on the sidewalk. Is the man wearing a helmet? No he does not have a helmet on. ... Are there any people nearby? Yes there's a woman walking behind him.

t rounds of history
(concatenated)



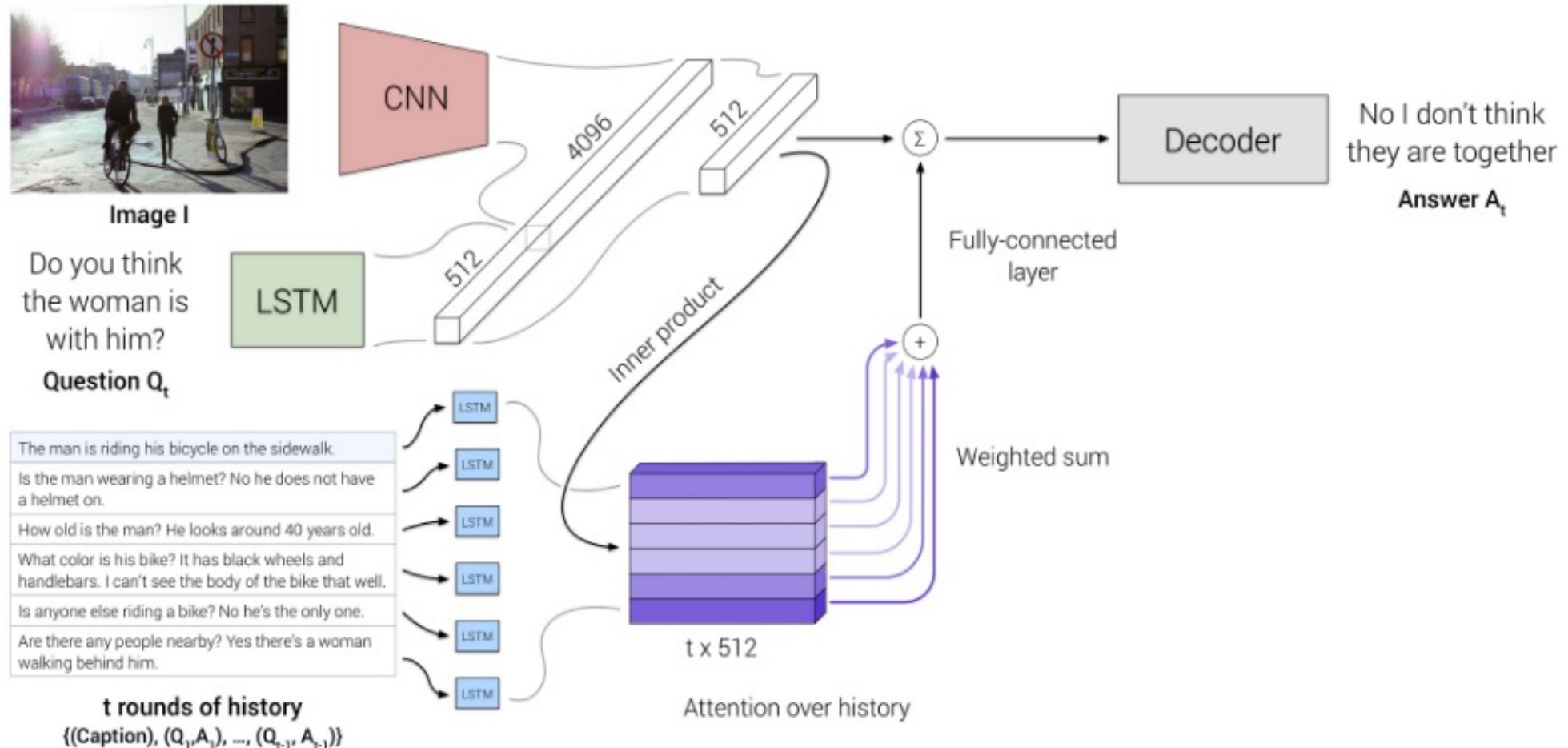
Visual Dialog – Hierarchical Recurrent Encoder

- Each QA-pairs in history H separately encoded by the LSTM



Visual Dialog – Hierarchical Recurrent Encoder

- Compute inner product of question vector with each history vector to get scores over previous rounds, which are fed to a SoftMax to get attention-over-history probabilities.



Improving Visual Dialog (and few examples)

- Generates more diverse, image relevant and fluent dialog.

Image + Caption	GT dialog (Das et al., 2017a)	Q-BOT-A-BOT dialog	Diverse-Q-BOT-A-BOT dialog
 a cutting board with carrots next to a cell phone.	what color is the cutting board? blue how big is it about the size of a regular laptop screen how many carrots are there ? 4 are they the orange kind ? 2 are orange , 2 are white what kind of cell phone is it ? a samsung can you see any text on the screen ? a recipe for UNK UNK soup do you see the room ? just the counter are there people ? not visible what color is the counter ? white	is the photo in color ? no how many people do you see ? no people is this a close up ? no is this a close up ? zebra is is this a close up ? no is this a close up ? no is this a close up ? no is this a close up ? no is this a close up ? no is this a close up ? no is this a close up ? no	is the photo in color ? 1 picture is in color is the photo close up ? black and white is the photo in color ? no it 's black and white is it a close up ? no it 's a close up of the cake can you tell what brand it is ? no i ca n't see the brand can you tell if it 's a male or female ? no i ca n't tell can you tell if it 's a male or female ? no i ca n't tell is this in a kitchen ? nope it 's in a home are there any utensils ? no are there any utensils ? nope
 many brown and black cows standing in the grass	is the field large ? yes is the grass green or brown ? green can you see a barn ? no can you see a fence ? no can you see a water supply ? yes can you see any people ? no can you see trees ? yes can you see mountains in the background ? no is the grass tall or short ? short	what color are the cows ? they are brown and white how many cows are there ? 4 what color are the cows ? they are brown and white can you see the sky ? it is not visible do you see any trees ? i do not see any trees can you see the sky ? green sky are there any clouds ? the sky is not visible how many cows are there ? 2 what color are the cows ? the cows are brown and white what color are the cows ? they are brown	do you see any people ? 1 do you see a fence ? a small 1 how many cows are there ? 5 what color are the cows ? black and white can you see the sky ? some can you see the sky ? nope are there any other animals ? not that i can see can you see the sky ? not at all are there any trees ? in the background can you see the sky ? not at all
 many elephants standing in an enclosure near a building	what type of building ? may be part of a zoo how many elephants ? about 4 is it sunny ? clear day , yes any trees ? yes what color is the building ? dark grey are there other animals ? no are there any people ? yes what are they wearing ? shorts and shirts how many ? 3	how many elephants are there ? ca n't tell what color are the elephants ? ca n't tell how many elephants are there ? ca n't tell what color is the fence ? i ca n't tell can you see the sky ? i ca n't see the sky are there any trees ? just a few does this look like a zoo ? 1 of them is are there any people ? nothing in the picture how many elephants are there ? just 1 can you see the sky ? just a little	how many elephants are there ? 1 any people ? ca n't see any male or female ? not sure what is the elephant doing ? just standing there male or female ? there are 2 males and 1 female can you see the sky ? 1 of it what color is the building ? just a black and white picture do you see any trees ? there are trees in the background are there any other animals ? red and white do you see any trees ? i see some in the background
 a reflection of the tub is in the bathroom mirror along with the sink , you can see all the room	is the photo in color ? yes is there humans in the photo ? no is there any animals in the photo ? no what color are the walls ? tan what color is the sink ? white is there any towels ? no is there any pictures on the wall ? no what color is the tub ? white can you see the floor ? no	is the photo in color ? 5 of them are are there any people ? it 's a close up so i ca n't see any people what color is the sink ? about the same color as the toilet do you see a mirror ? no what color is the sink ? yes it is white what color is the sink ? yes it is white what shape is the mirror ? yes , it 's a rectangle are there any towels ? he 's on a towel rack are there any towels ? i do n't see any towels are there any towels ? i do n't see any towels	is the photo in color ? no is there any people ? 5 what color walls ? i ca n't see the walls is there a mirror ? yes is there a mirror ? yes is there a mirror ? yes are the walls seen ? no what color are the walls ? no walls what is the floor made of ? yes what color is the floor ? i ca n't see the floor

Many more tasks at CV+NLP

➤ Vision-Language Navigation (VLN)

Instruction

Turn right and head towards the *kitchen*. Then turn left, pass a *table* and enter the *hallway*. Walk down the hallway and turn into the *entry way* to your right *without doors*. Stop in front of the *toilet*.

-  Initial Position
-  Target Position
-  Demonstration Path A
-  Executed Path B
-  Executed Path C

Local visual scene



Global trajectories in top-down view

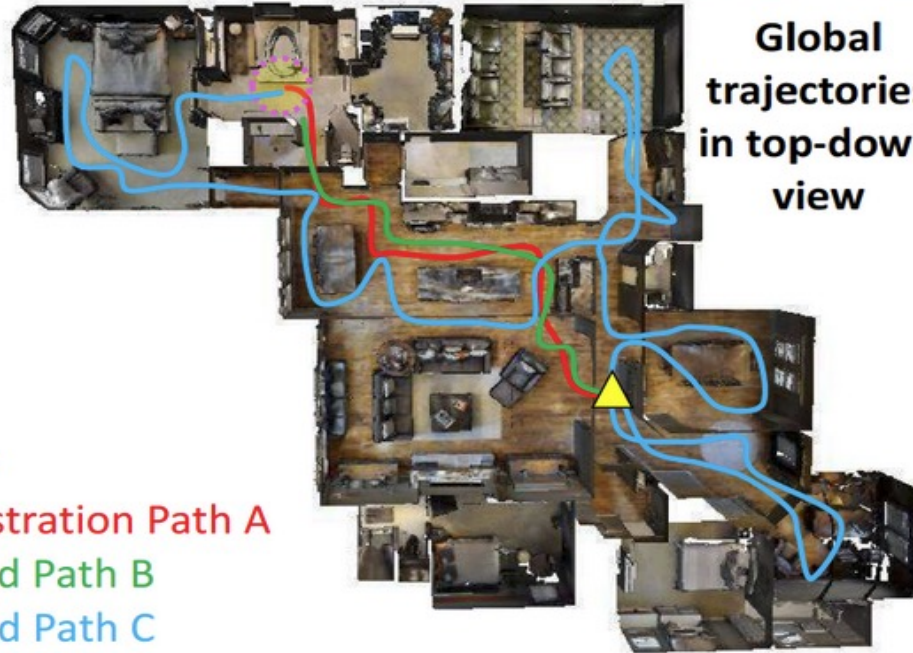


Figure 1: Demonstration of the VLN task. The instruction, the local visual scene, and the global trajectories in a top-down view. The agent does not have access to the top-down view. Path A is the demonstration path following the instruction. Path B and C represent two different paths executed by the agent. Figure credit: Wang et al. (2019).

Many more tasks at CV+NLP

- Visual grounding for referring expressions
 - Grounding an utterance to refer to something or someone in image

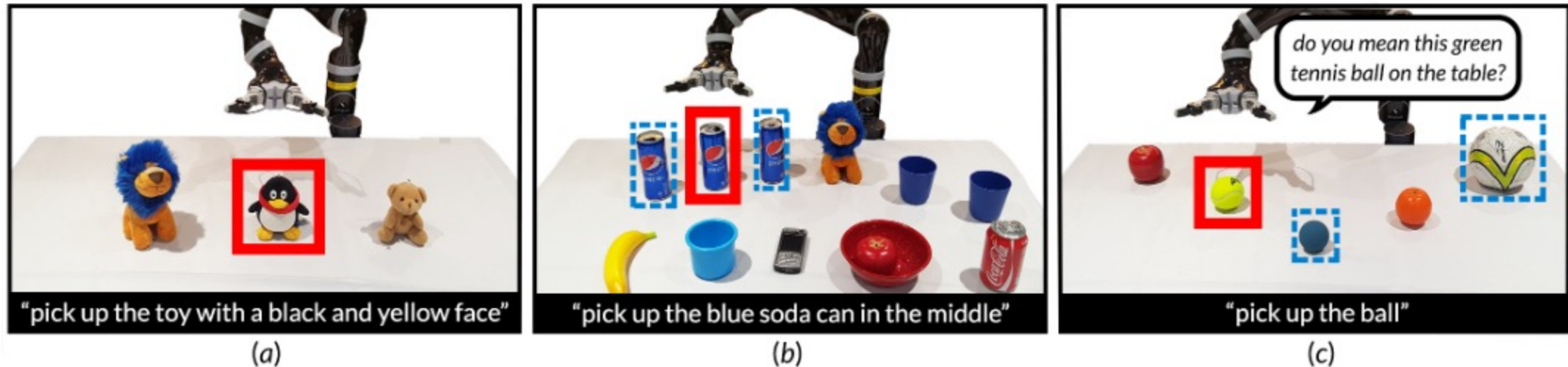


Fig. 1: Interactive visual grounding of referring expressions. (a) Ground self-referential expressions. (b) Ground relational expressions. (c) Ask questions to resolve ambiguity. Red boxes indicate referred objects. Blue dashed boxes indicate candidate objects. See also the accompanying video at <http://bit.ly/INGRESSvid>.

- Let's see video in action: <http://bit.ly/INGRESSvid>

Many more tasks at CV+NLP

➤ Fun stuff: Comic books

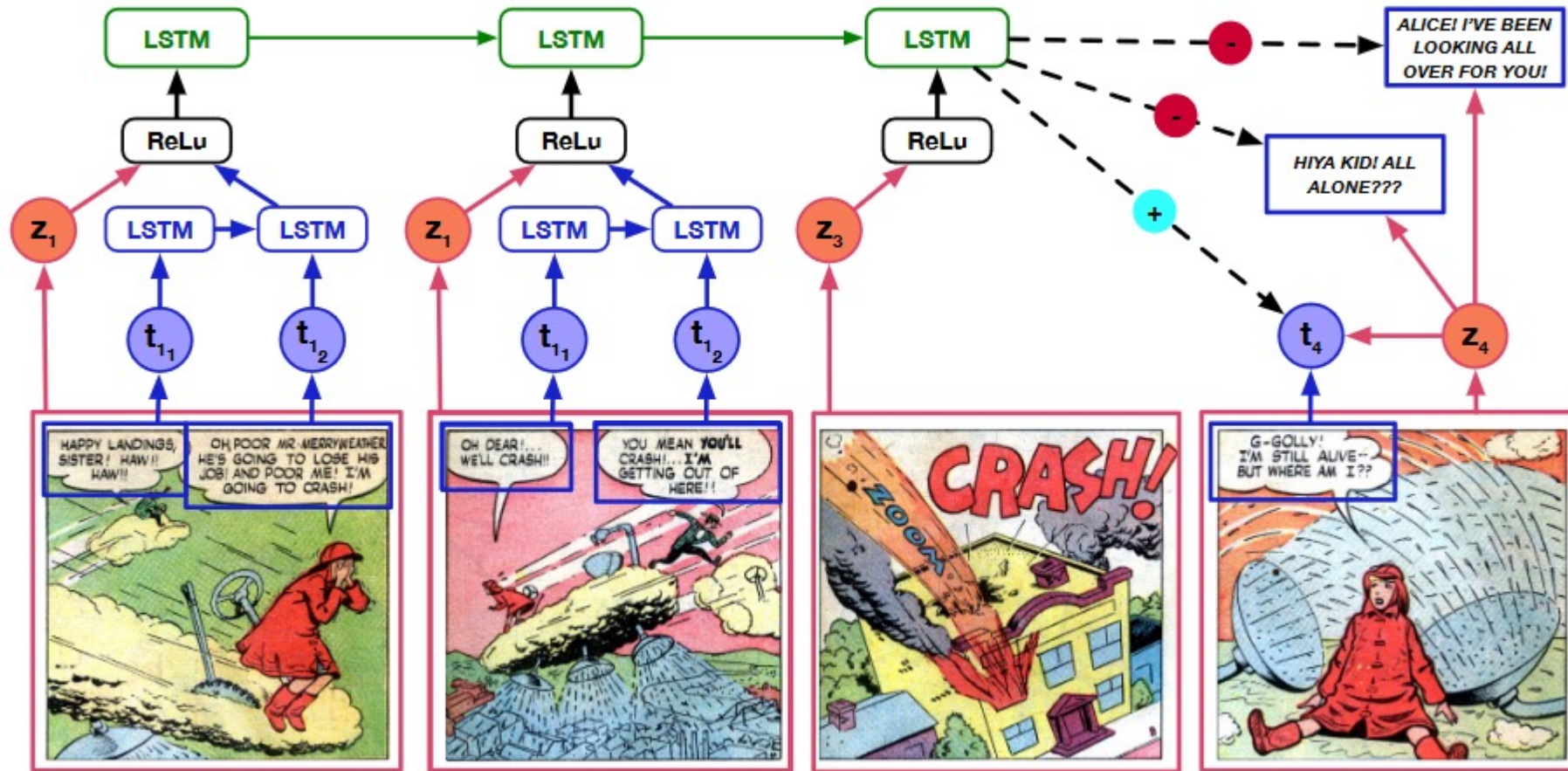
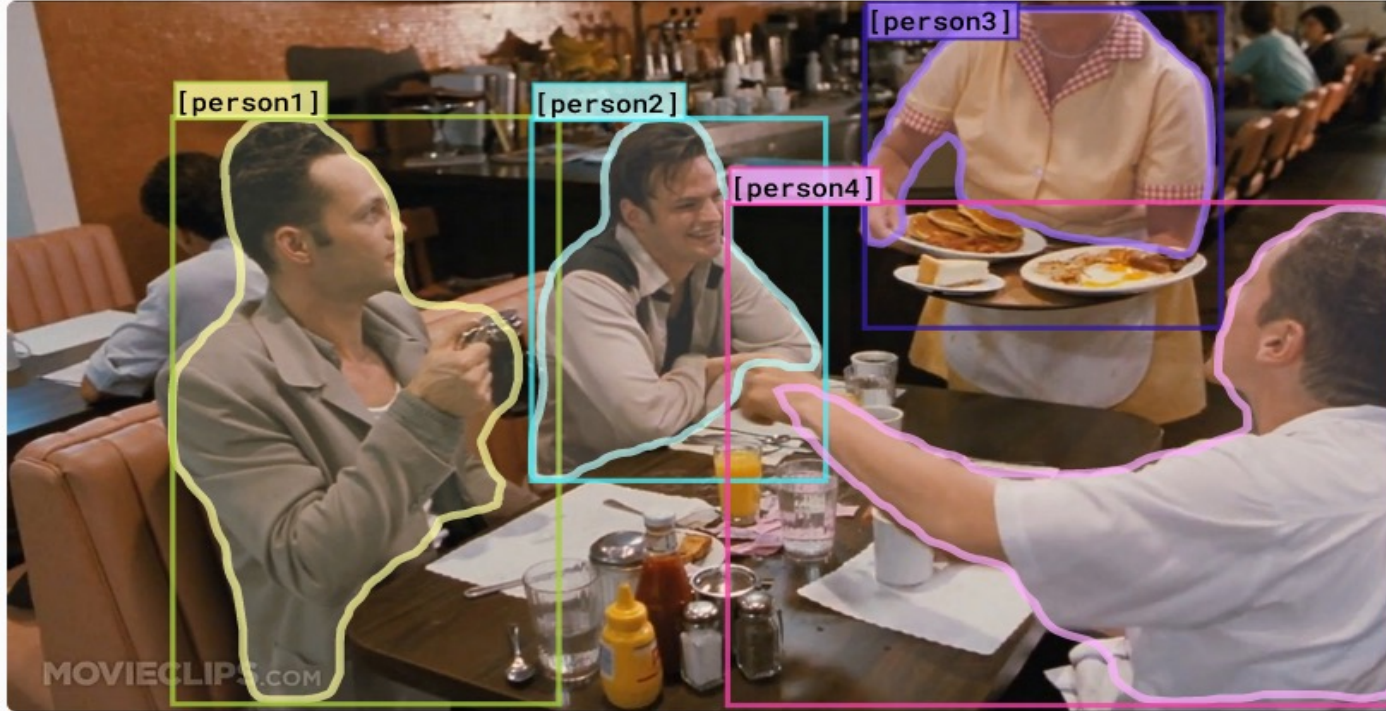


Figure 5. The image-text architecture applied to an instance of the *text cloze* task. Pretrained image features are combined with learned text features in a hierarchical LSTM architecture to form a context representation, which is then used to score text candidates.

Many more tasks at CV+NLP

➤ From Recognition to Cognition: Visual Commonsense Reasoning



hide all

show all

[person1]

[person2]

[person3]

[person4]

more objects »

Why is [person4] pointing at [person1]?

- a) He is telling [person3] that [person1] ordered the pancakes.
- b) He just told a joke.
- c) He is feeling accusatory towards [person1].
- d) He is giving [person1] directions.

Rationale: I think so because...

- a) [person1] has the pancakes in front of him.
- b) [person4] is taking everyone's order and asked for clarification.
- c) [person3] is looking at the pancakes both she and [person2] are smiling slightly.
- d) [person3] is delivering food to the table, and she might not know whose order is whose.

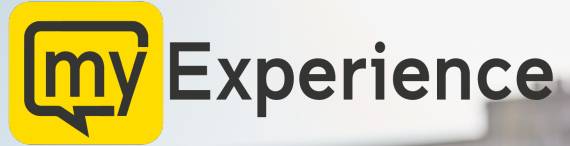
Summary

- Vision + Language: Help us to understand human brain functioning better
- Visual Turing Test for modern AI systems
- A step towards Artificial General Intelligence (AGI)
- Variety of applications:
 - Helping visually-impaired people
 - Early child education
 - Personal assistants
 - Robot navigation
 - Video surveillance systems
 - Search engines



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Questions?

If you feel fascinated and want to get involved in research, feel free to reach out at sonit.singh@unsw.edu.au